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VICINITY MAP

N.T.S.

PROJECT DESCRIPTION:

NATURAL GAS WELL PAD AND 1,180-FOOT ± ACCESS DRIVE CONSTRUCTION.
THE PROJECT LIMITS OF CONSTRUCTION TOTAL 5.65± ACRES; TOTAL CLEARING LIMITS 10.35± ACRES

CONSTRUCTION SEQUENCE:

1. PRIOR TO STARTING ANY EARTH DISTURBANCE ACTIVITIES, THE CONTRACTOR SHALL CONTACT THE APPLICABLE ONE CALL SYSTEM FOR LOCATION OF ALL UTILITIES WITHIN THE VICINITY OF THE PROJECT.
2. INSTALL STABILIZED ACCESS DRIVE ENTRANCE, PER DETAIL SHEET 15.
3. CLEAR & GRUB FOR CONSTRUCTION ENTRANCE AND ACCESS DRIVE. ANY TIMBER REMOVED WITHIN THE CLEARING LIMITS SHALL BE TAKEN CARE OF UNDER THE DIRECTION OF ASCENT RESOURCES. NON-DECOMPOSING DEBRIS SHALL BE REMOVED AND DISPOSED OF OFF-SITE IN AN APPROVED MANNER ACCORDING TO ALL STATE AND LOCAL REGULATIONS.
4. INSTALL PERIMETER EROSION & SEDIMENT CONTROL MEASURES, AS SPECIFIED HEREIN.
5. STRIP AND STOCKPILE TOPSOIL WITHIN CONSTRUCTION LIMITS.
6. ROUGH GRADING.
7. FINE GRADING AND PAVEMENT CONSTRUCTION.
8. PLACE 4" OF TOPSOIL IN AREAS DISTURBED BY CONSTRUCTION. SEED AND MULCH PER CMS ITEM 659. PROVIDE SEEDING & MULCHING WITH TACK IN AREAS INDICATED. PROVIDE TEMPORARY SEEDING OF REMAINING TOPSOIL STOCKPILES (TO BE USED IN SITE RECLAMATION). (NOTE: TOPSOIL AND TRADITIONAL SEEDING/MULCHING METHODS CAN BE OMITTED IN AREAS WHICH ARE NOT CONDUCTIVE. STABILIZATION OF THESE AREAS TO BE ACHIEVED BY ALTERNATIVE TECHNIQUES APPROVED BY THE ENGINEER, SUCH AS HYDRO SEEDING/MULCHING, EROSION CONTROL BLANKETS, ETC.). RE-SPREADING OF TOPSOIL SHALL BE DELAYED DURING WINTER MONTHS UNTIL WEATHER IS SUITABLE FOR SUCH OPERATIONS, AND THE PROMPT APPLICATION OF PERMANENT SEEDING IS FEASIBLE.
9. DRILLING/COMPLETIONS OPERATIONS.
10. REMOVE EROSION & SEDIMENT CONTROL MEASURES ONCE SITE HAS STABILIZED (70% GROWTH)



WALDO MNR GR WELL PAD

SURFACE LOCATION

(NAD 83) 40.174505 N; 81.448581 W

(NAD 27) 40.174435 N; 81.448743 W

ACCESS LOCATION

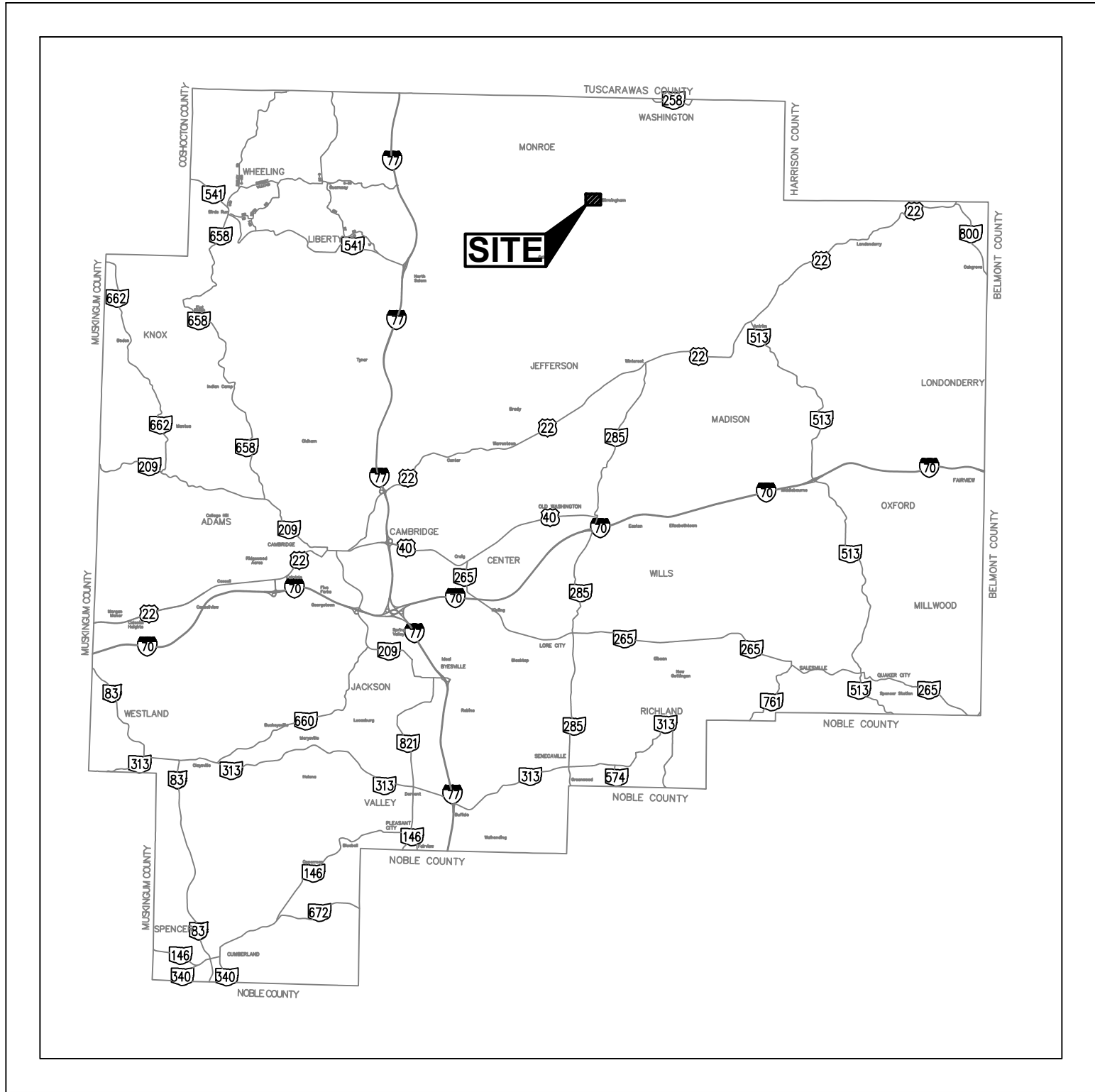
(NAD 83) 40.171476 N; 81.445555 W

(NAD 27) 40.171406 N; 81.445717 W

MONROE TOWNSHIP GUERNSEY COUNTY, OHIO (SECTION 11, TOWNSHIP 4, RANGE 2)

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LOCATION MAP

N.T.S.

LAND OWNER:

MILLER WALDO C II & CARLA J
PID: 260000124000 & 260000122000

DEVELOPER:

ASCENT RESOURCES, LLC
1000 UTICA WAY
CAMBRIDGE, OH 43725
FIELD SUPERVISOR: ASHLEY COWAN
P: 254.485.6226
E: ASHLEY.COWAN@ASCENTRESOURCES.COM
EMERGENCY CONTACT: MATT MROCZKOWSKI
P: 614.896.0362
E: MATT.MROCZKOWSKI@ASCENTRESOURCES.COM

ENGINEER:

CESO, INC.
2800 CORPORATE EXCHANGE DRIVE, SUITE 160
COLUMBUS, OHIO 43231
CONTACT: JARROD MAHAFFEY
P: 614.794.7080
E: MAHAFFEY@CESOINC.COM

GEO-TECHNICAL ASSOCIATES, INC.
2292 INDUSTRIAL HIGHWAY
YORK, PENNSYLVANIA 17402
CONTACT: JON RAAB
P: 717.318.5451
E: JRAAB@GTAENG.COM



REVISION DESCRIPTION

NO. DATE



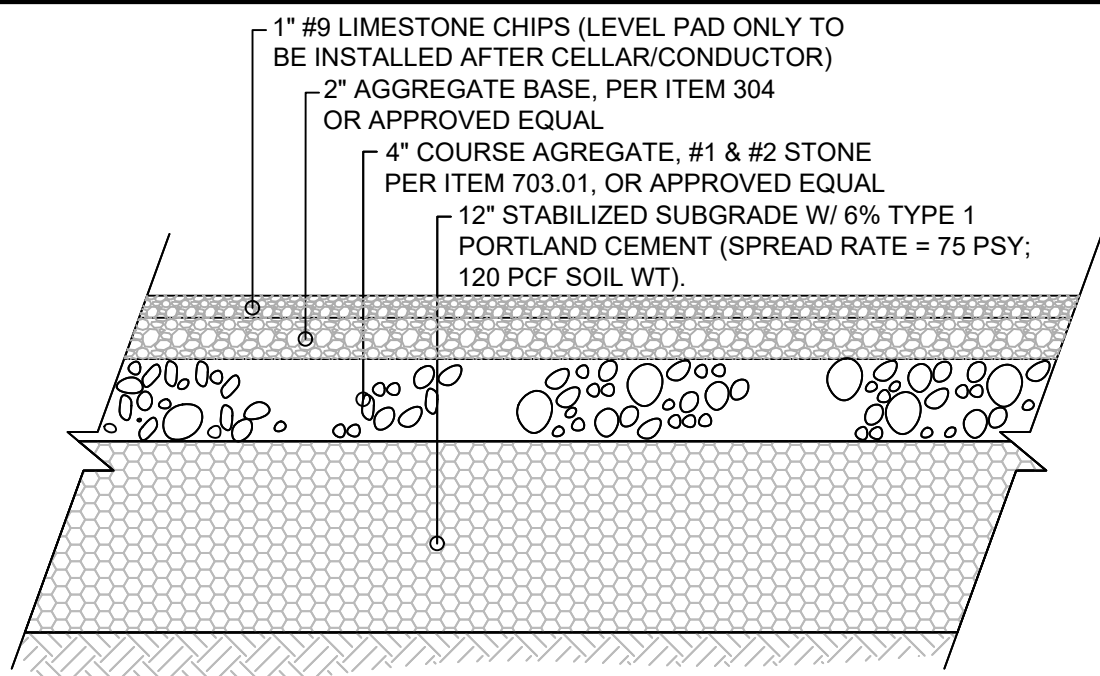
WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

TITLE SHEET

ISSUE:	CONST. DOCUMENTS
DATE:	08.14.2018
JOB NO.:	755215
DESIGN:	JMR
DRAWN:	JMR
CHECKED:	JRM
SHEET NO.	1

ENGINEER'S ESTIMATE OF QUANTITIES		
DESCRIPTION	QUANTITY	UNITS
SITE CLEARING AND EROSION & SEDIMENT CONTROL		
CLEARING & GRUBBING	8.40	AC
TREE REMOVAL, DENSE	1.95	AC
FILTER SOCK, 18"	2,505	LF
FILTER SOCK, 24"	835	LF
FILTER SOCK, 32"	685	LF
VEGETATIVE SWALES	2,260	LF
ROCK CHECK DAM, TYPE A	19	EA
ROCK CHANNEL PROTECTION, TYPE D	100	CY
STABILIZED CONSTRUCTION ENTRANCE	1	EA
SEEDING & MULCHING W/ TACK	1.85	AC
SEEDING & MULCHING	4.75	AC
12" STORM PIPE CULVERT	145	LF
12" IN-LINE DRAIN (INC. RISER PIPE, GRATE, ETC.)	1	LS
ACCESS DRIVE CONSTRUCTION		
EARTHWORK, EXCAVATION	690	CY
EARTHWORK, EMBANKMENT	1,670	CY
TOPSOIL STRIPPED AND STOCKPILED, ITEM 651	975	CY
PLACING STOCKPILED TOPSOIL, ITEM 652	315	CY
12" STABILIZED SUBGRADE (INC. CEMENT)	105	TON
AGGREGATE BASE, ITEM 304	155	CY
COARSE AGGREGATE, ODOT #1 & #2 STONE, ITEM 703.01	310	CY
AGGREGATE BASE, ITEM 304 (STABILIZED CONSTRUCTION ENTRANCE)	10	CY
COARSE AGGREGATE, ODOT #1 & #2 STONE, ITEM 703.01 (STABILIZED CONSTRUCTION ENTRANCE)	50	CY
WELL PAD CONSTRUCTION		
EARTHWORK, EXCAVATION	23,130	CY
EARTHWORK, EMBANKMENT	22,090	CY
TOPSOIL STOCKPILED	3,580	CY
PLACING STOCKPILED TOPSOIL	680	CY
TOE KEY EXCAVATION	2,585	CY
12" STABILIZED SUBGRADE (INC. CEMENT)	575	TON
AGGREGATE BASE, ITEM 304	855	CY
COARSE AGGREGATE, ODOT #1 & #2 STONE, ITEM 703.01	1,705	CY
#9 LIMESTONE CHIPS	185	CY
4" UNDERDRAIN (INC. BEDDING AND GEOTEXTILE)	1,345	LF
WELL PAD OUTLET	2	EA

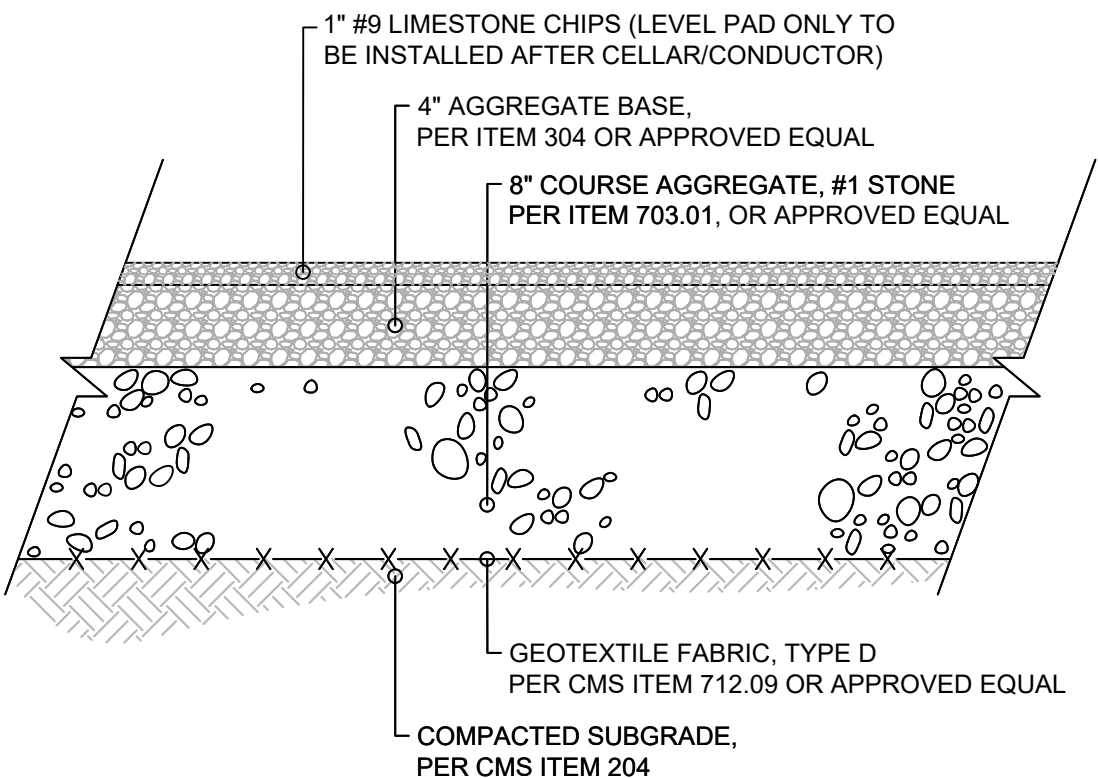


NOTES:

1. CEMENT STABILIZED SUBGRADE, AS DIRECTED - DURING CONSTRUCTION, THE ENGINEER WILL OBTAIN BULK SAMPLES OF SUBGRADE SOILS, AND PREPARE CEMENT MIX DESIGNS NECESSARY TO ACHIEVE 150-200 PSI STRENGTH AT 7-DAY CURE.

FOR BIDDING PURPOSES, THE ASSUMED SPREAD RATE SHALL BE 6.0% CEMENT CONTENT BY VOLUME. THE FINAL MIX DESIGN WILL BE PROVIDED TO THE STABILIZATION CONTRACTOR TWO DAYS PRIOR TO STARTING WORK.

WELL PAD AND ACCESS DRIVE STABILIZED AGGREGATE SECTION

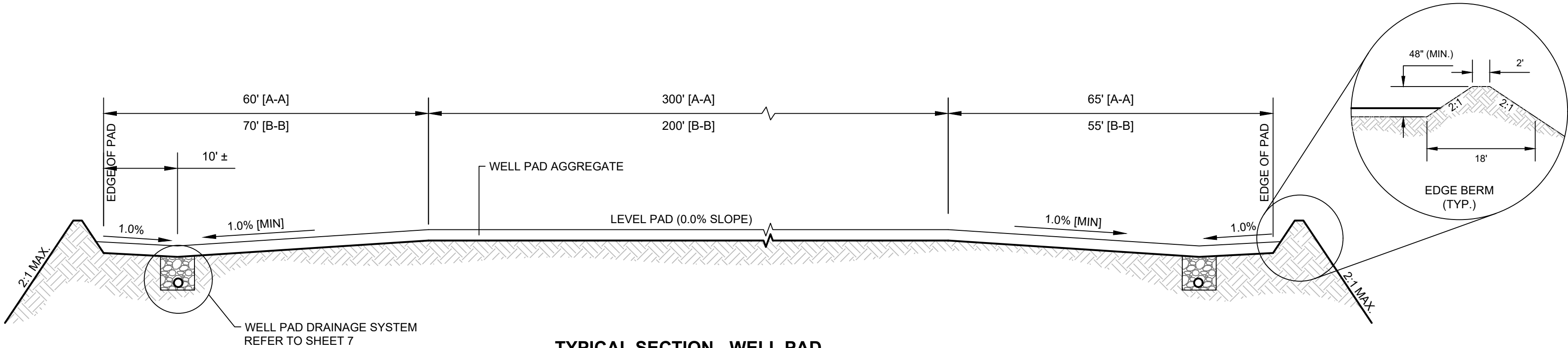
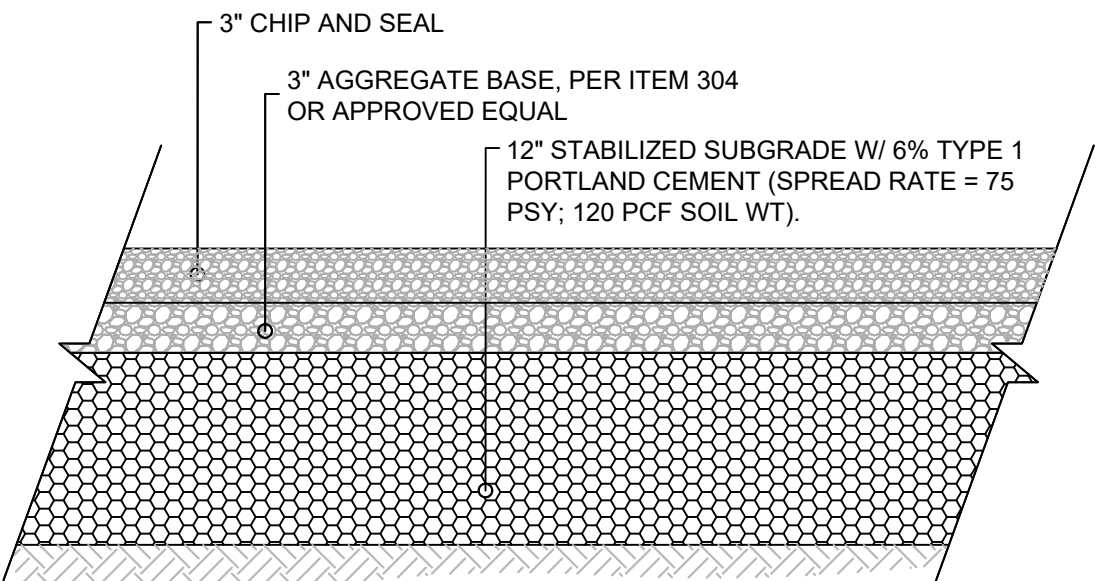


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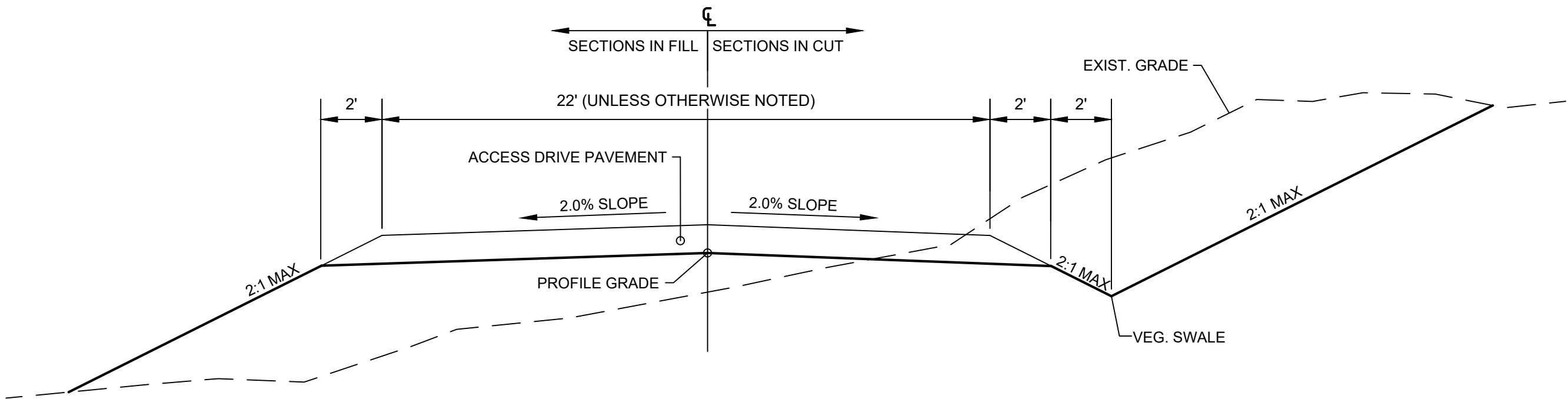
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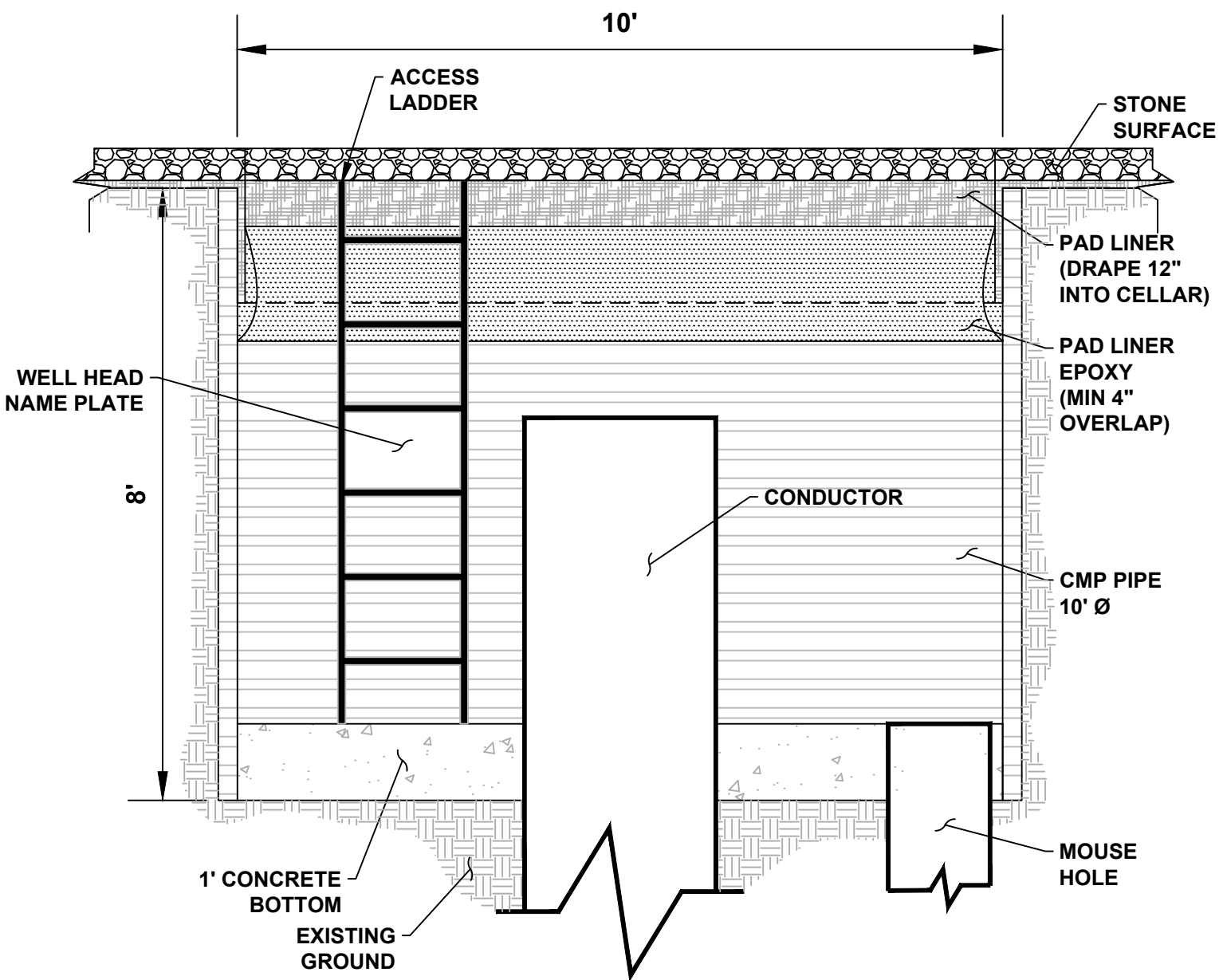
ALTERNATE WELL PAD & ACCESS DRIVE AGGREGATE SECTION



TYPICAL SECTION - WELL PAD



TYPICAL SECTION - ACCESS DRIVE



CELLAR DETAIL

[illegible]

WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

ESTIMATE OF QUANTITIES & TYPICAL SECTIONS

ISSUE: CONST. DOCUMENTS	
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EXISTING	PROPOSED
R/W	WELL SITE BOUNDARY
PROPERTY LINE	
1000	
MAJOR CONTOUR	
1002	
MINOR CONTOUR	
PAVEMENT	
TREE LINE	
STREAM	
WETLAND	
SOIL LIMITS	
CONTROL POINT	
WATER WELL	
TREE	
PLUGGED AND ABANDONED WELL ID #2208	

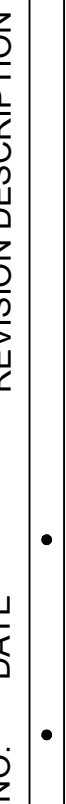
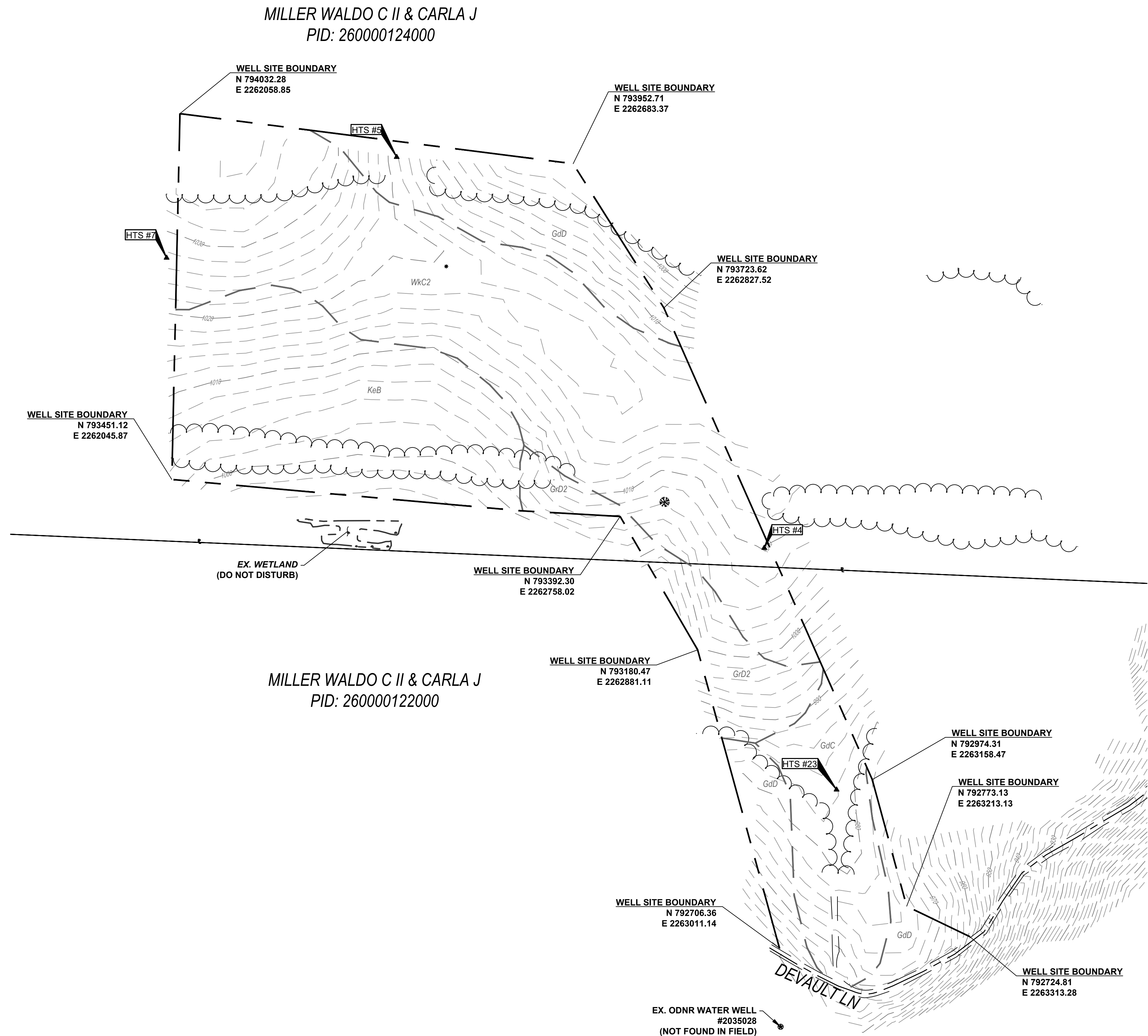
SOIL CLASSIFICATION TABLE	
SOIL NAME	PERCENT (%)
GILPIN SILT LOAM (GdC)	10.0
GILPIN SILT LOAM (GdD)	17.9
GUERNSEY SILT LOAM (GrD2)	6.0
KEENE SILT LOAM (KeB)	23.1
WESTMORE SILT LOAM (WkC2)	43.0

CONTROL POINTS TABLE (STATE PLANE NAD 83)			
POINT NAME	NORTHING*	EASTING*	ELEVATION**
HTS #4	793342.24	2262986.85	1010.20
HTS #5	793963.31	2262403.05	1018.55
HTS #7	793803.22	2262037.62	1027.44
HTS #23	792958.29	2263101.79	982.61

*GPS OBSERVATION OHIO STATE PLANE SOUTH ZONE NAD 83
** GPS OBSERVATION NAVD 88, GEOID12

1. A REVIEW OF THE ODNR OHIO MINE VIEWER SHOWED NO MINING ACTIVITY WITHIN THE WELL SITE BOUNDARY AND 100' BEYOND SEE SHEET 4 FOR DETAILS.
2. PROPERTY LINES ARE SHOWN FOR GRAPHICAL PURPOSES ONLY AND ARE BASED ON RECORD INFORMATION. NO BOUNDARY SURVEY WAS PERFORMED.
3. COORDINATES BASED UPON GPS OBSERVATION OHIO STATE PLANE SOUTH ZONE NAD83, CORS2011. ELEVATIONS BASED UPON GPS OBSERVATION NAVD88, GEOID12.
4. A REVIEW OF OHIO DEPARTMENT OF NATURAL RESOURCES RECORDS FOR WATER, OIL, AND GAS WELLS WAS PERFORMED WITHIN THE WELL SITE BOUNDARY AND 100' BEYOND. THAT REVIEW IS REFLECTED ON THIS SHEET.

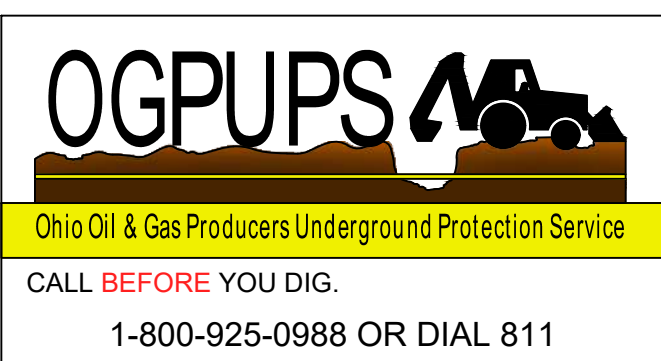
THIS IS TO CERTIFY THAT THE INFORMATION SHOWN ON THIS/THESE PLAN(S)
ACCURATELY DEPICTS FIELD CONDITIONS BASED ON TOPOGRAPHIC SURVEY
CONDUCTED IN JUNE 2018.



WALDO MNR GR WELL PAD

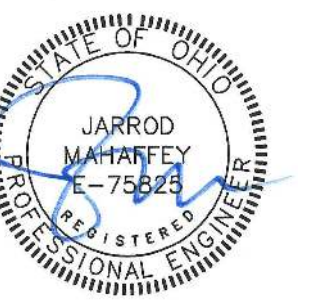
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COAL MINING - ABANDONED UNDERGROUND MINE EXTENTS: GY-127

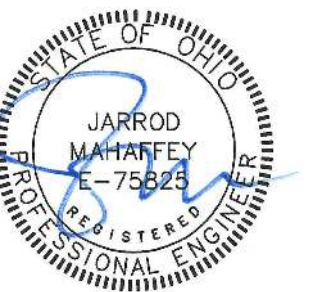
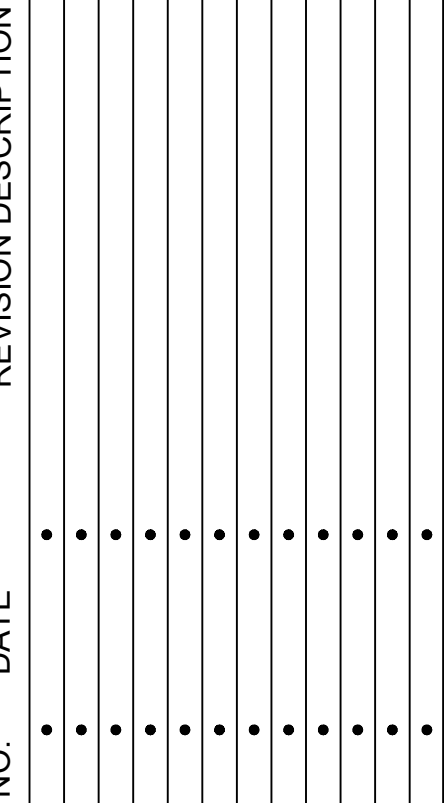


EXISTING MINES MAP

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08.14.2018	
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SIGN:	JMR
AWN:	JMR
ECKED:	JRM
SHEET NO.	
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A north arrow pointing upwards, labeled with a large 'N'. Below the arrow is a circular symbol with a crosshair, divided into four quadrants. At the bottom is a graphic scale bar with alternating black and white segments. The scale is marked with 100, 0, 100, and 200. Below the scale bar, the text reads: GRAPHIC SCALE (IN FEET) 1 inch = 100 ft.



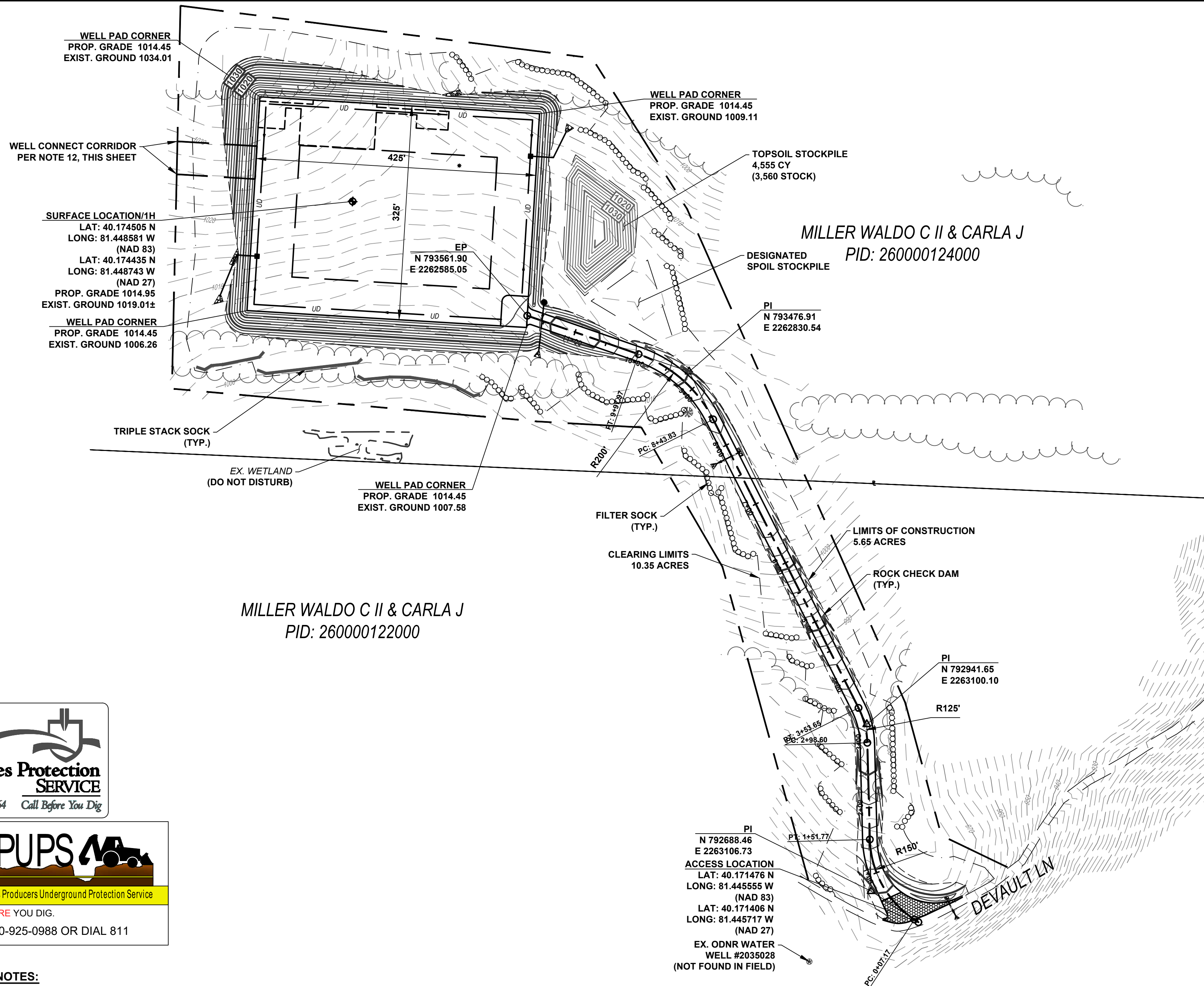
ASCENT
RESOURCES

WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

SITE DISTURBANCE PLAN

ISSUE:	
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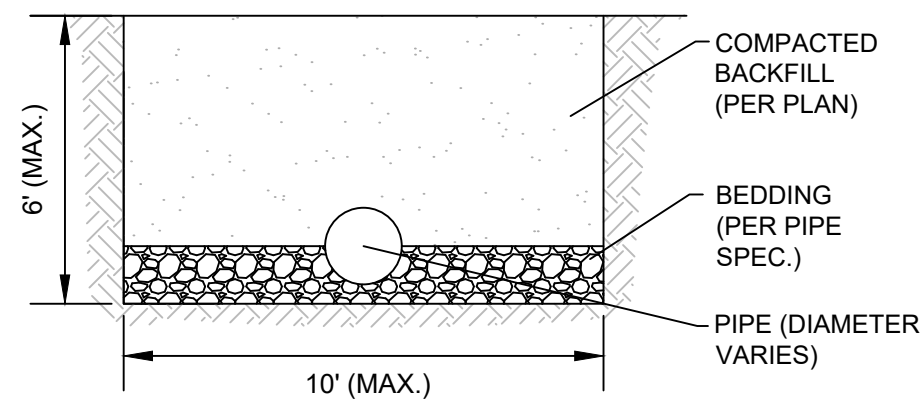
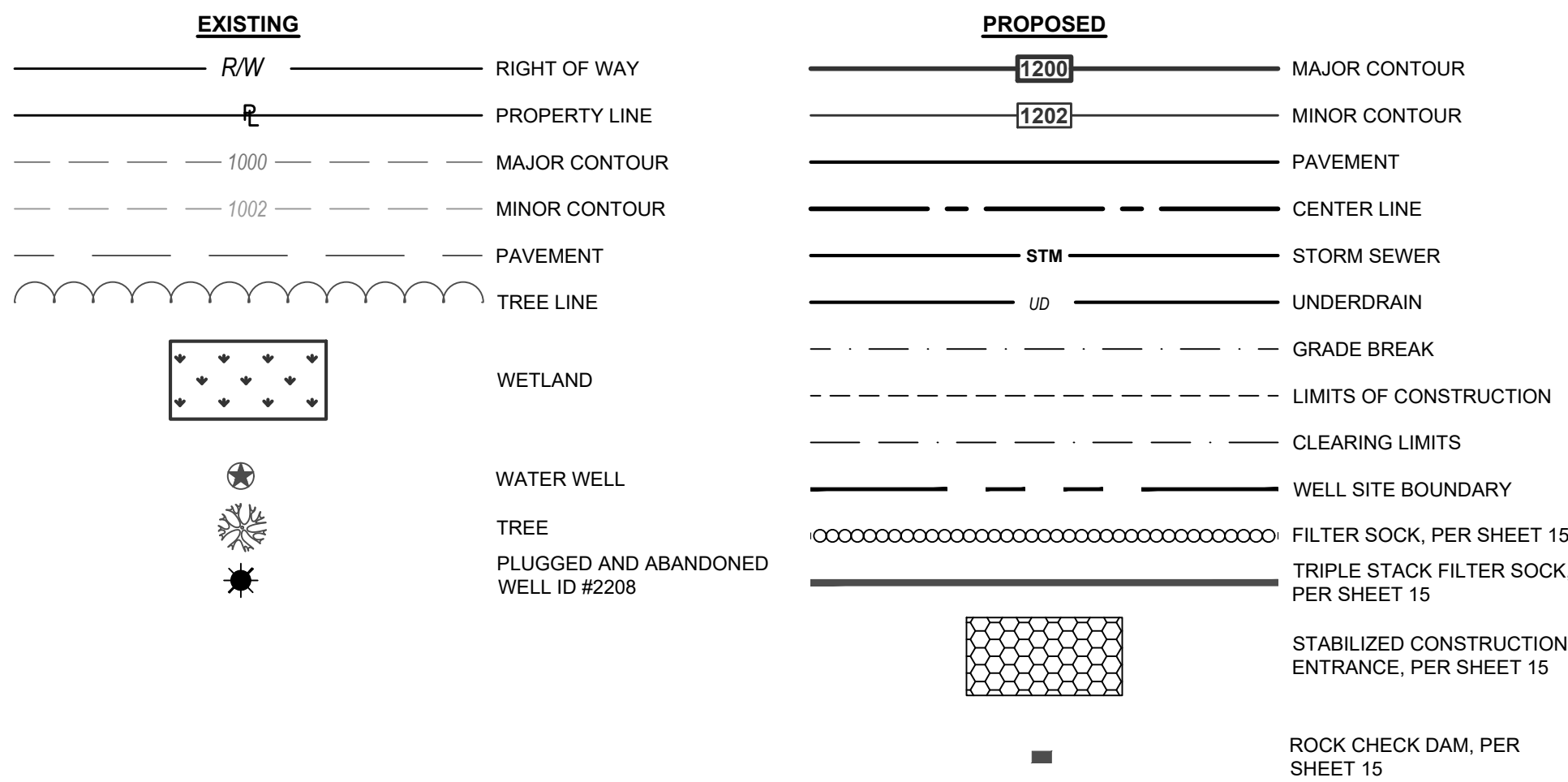


GENERAL NOTES:

1. ALL WORK AND MATERIALS SHALL COMPLY WITH ALL LOCAL REGULATIONS AND CODES AND O.S.H.A. STANDARDS. THE CONTRACTOR SHALL OBTAIN FINAL PERMITTING AND APPROVAL/INSPECTIONS AS REQUIRED FROM THE LOCALITY.
2. THIS PLAN DOES PURPORT TO BE A COMPREHENSIVE REPRESENTATION OF EXISTING UTILITIES IN THE PROJECT AREA. UTILITIES SHOWN ARE BEING PROVIDED FOR INFORMATIONAL PURPOSES ONLY, BASED ON SURFACE FEATURES OBSERVED IN THE FIELD. CESO, INC. MAKES NO GUARANTEE TO THEIR ACCURACY AND/OR COMPLETENESS. FOURTY-EIGHT (48) HOURS BEFORE DIGGING IS TO COMMENCE, THE CONTRACTOR SHALL NOTIFY THE OHIO UTILITIES PROTECTION SERVICE (OUPS) AT 1-800-362-2764. THE CONTRACTOR SHALL CALL THE OIL & GAS PRODUCERS UNDERGROUND PROTECTION SERVICE (OGPUPS) AT 1-800-925-0988. THE CONTRACTOR SHALL ALSO NOTIFY ALL OTHER AGENCIES WHICH MIGHT HAVE UNDERGROUND UTILITIES IN THE PROJECT AREA AND ARE NON-MEMBERS OF OUPS.
3. THE CONTRACTOR SHALL MAINTAIN ALL EROSION & SEDIMENT CONTROL MEASURES LISTED ON SHEETS 12-15. ALL LAND DISTURBING ACTIVITIES ARE SUBJECT TO INSPECTION BY THE OHIO ODNR TO DETERMINE COMPLIANCE WITH THEIR STANDARDS AND REGULATIONS.
4. STORM SEWERS SHALL BE HIGH-DENSITY POLYETHYLENE PIPE (HDPE) - AASHTO M 252, M 294, TYPE S. PIPE BEDDING AND BACKFILL SHALL CONSIST OF NATIVE MATERIALS, FREE OF ORGANIC AND ROCKS, MEETING USCS DESIGNATIONS OF CL OR ML. UNSUITABLE SOILS, SUCH AS CH OR MH TYPE SOILS ARE NOT PERMITTED. STORM SEWERS SHALL BE INSPECTED AND APPROVED PRIOR TO BACKFILLING.
5. ALL PLAN COORDINATES, UNLESS OTHERWISE NOTED ARE IN STATE PLANE COORDINATE SYSTEM (OHIO SOUTH ZONE), NAD83 AND ELEVATIONS BASED UPON GPS OBSERVATION NAVD88, GEOID12.
6. A REVIEW OF OHPO ONLINE DATABASE FOUND NO KNOWN HISTORICALLY SIGNIFICANT FEATURES WITHIN THE PROJECT VICINITY. HOWEVER, IF DURING THE COURSE OF CONSTRUCTION ARTIFACTS BELIEVING TO HAVE SIGNIFICANT HISTORICAL OR ARCHAEOLOGICAL SIGNIFICANCE ARE ENCOUNTERED, THE CONTRACTOR SHALL HALT OPERATIONS AND CONTACT MATT MROCKZKOWSKI AT (614) 896-0362.
7. A REVIEW OF OEPA SOURCE WATER ASSESSMENT AND PROTECTION PROGRAM FOUND THAT ALL PORTIONS OF THE WELL SITE BOUNDARY & 100 FEET BEYOND ARE OUTSIDE OF THE SURFACE WATER 5-YEAR TIME OF TRAVEL AS WELL AS ANY EMERGENCY MANAGEMENT ZONE OF A PUBLIC WATER SYSTEM INTAKE.
8. A REVIEW OF THE NATIONAL FLOOD INSURANCE RATE MAP FOUND THAT ALL PORTIONS OF THE WELL SITE BOUNDARY & 100 FEET BEYOND ARE LOCATED OUTSIDE OF THE DELINEATED FEMA RISK ZONES & HAZARDOUS AREAS.
9. AN ENVIRONMENTAL ASSESSMENT FOR THE PROJECT AREA HAS BEEN PREPARED BY CESO, INC. DATED 07/13/2018. IN COORDINATION WITH USFWS, IT HAS BEEN DETERMINED THAT THERE ARE NO KNOWN CAPTURES OF NORTHERN LONG EARED BATS WITHIN THE PROJECT VICINITY. THERE IS ANTICIPATED TREE CLEARING ASSOCIATED WITH CONSTRUCTION. THEREFORE, TREE CLEARING IS PROHIBITED EXCEPT IN THE APPROVED CLEARING SEASON OF OCTOBER 1 THROUGH MARCH 31.

ADDITIONALLY, A REVIEW OF THE WELL SITE BOUNDARY AREA AND 100' BEYOND HAS LOCATED STREAMS AND/OR WETLANDS AS SHOWN ON THIS PLAN.
10. A GEOTECHNICAL EXPLORATION AND ANALYSES WAS PERFORMED BY GTA, DATED 07/13/2018 PLEASE REFERENCE THIS DOCUMENT IN REGARDS TO MATERIALS AND EARTHWORK RECOMMENDATIONS.
11. IN LOCATIONS WHERE EARTHWORK TRANSITIONS FROM CUT-TO-FILL WITHIN THE WELL PAD LIMITS, THE CONTRACTOR SHALL UNDERCUT THE EXISTING GRADE (40' WIDE X 2' DEEP) AND RECOMPACT WITH PROPOSED EMBANKMENT LIFTS.
12. WELL CONNECT CORRIDOR SHALL BE UTILIZED FOR LOCATION OF MIDSTREAM CONNECTION. ALL EXCAVATION WITHIN THE WELL CONNECT CORRIDOR NEED TO BE LIMITED TO NO GREATER THAN 6' IN DEPTH & 10' IN WIDTH. NO OTHER DEEP EXCAVATIONS ARE PERMITTED WITHIN THE WELL SITE BOUNDARY ONCE SITE HAS REACHED FINAL GRADE.

LEGEND



WELL CONNECT TRENCH DETAIL

- NOTES:
1. ALL EXCAVATIONS MUST STAY WITHIN THE WELL CONNECT CORRIDOR SHOWN ON THIS SHEET
 2. ALL EXCAVATIONS NEED TO BE PERFORMED PERPENDICULAR TO ENGINEERED SLOPES & TOE KEYS
 3. IF ENCOUNTERED ALL BENCH & TOE KEY DRAINS MUST BE REPLACED IN KIND
 4. BACKFILL MUST BE COMPACTED IN LIFTS NO GREATER THAN 9 INCHES AT A TIME
 5. ALL EXCAVATIONS MUST BE BACKFILLED WITHIN 24 HOURS

WELL COORDINATE TABLE				
	NAD 83		NAD 27	
WELL #	LATITUDE	LONGITUDE	LATITUDE	LONGITUDE
TH/1H	40.174505 N	81.448581 W	40.174435 N	81.448743 W

EARTHWORK SUB-SUMMARY		
DESCRIPTION	QUANTITY	UNITS
ACCESS DRIVE		
RAW EXCAVATION (EXIST. SUBSOIL TO DRIVE SUBGRADE)	690	CY
RAW EMBANKMENT (EXIST. SUBSOIL TO DRIVE SUBGRADE)	1,670	CY
DRIVE EARTHWORK BALANCE (RAW CUT TO FILL)	980 [IMPORT]	CY
TOPSOIL STOCKPILED (6" AVG.)	975	CY
PLACING STOCKPILED TOPSOIL (4")	315	CY
EXCESS TOPSOIL	660	CY
WELL PAD		
RAW EXCAVATION (EXIST. SUBSOIL TO PAD SUBGRADE)	23,130	CY
RAW EMBANKMENT (EXIST. SUBSOIL TO PAD SUBGRADE)	22,090	CY
PAD EARTHWORK BALANCE (RAW CUT TO FILL)	1,040 [SPOIL]	CY
TOPSOIL STOCKPILED (6" AVG.)	3,580	CY
PLACING STOCKPILED TOPSOIL (4")	680	CY
EXCESS TOPSOIL	2,900	CY
TOE KEYS & BONDING BENCHES		
RAW EXCAVATION (EXIST. SUBSOIL TO SUBGRADE)	2,585	CY
RAW EMBANKMENT (EXIST. SUBSOIL TO SUBGRADE)	2,585	CY
NET SITE EARTHWORK = 60 CY [SPOIL]		

 TOPSOIL DEPTH SUMMARY*			
B-01 = 5 IN.	B-02 = 7 IN.	B-03 = 6 IN.	B-04 = 6 IN.
B-05 = 3 IN.			

* DEPTHS INDICATED ARE FOR INFORMATIONAL PURPOSES ONLY. THE CONTRACTOR SHALL STRIP AND STOCKPILE ALL TOPSOIL WITHIN THE CONSTRUCTION LIMITS. SEE SHEET 16 FOR BORING LOCATIONS.

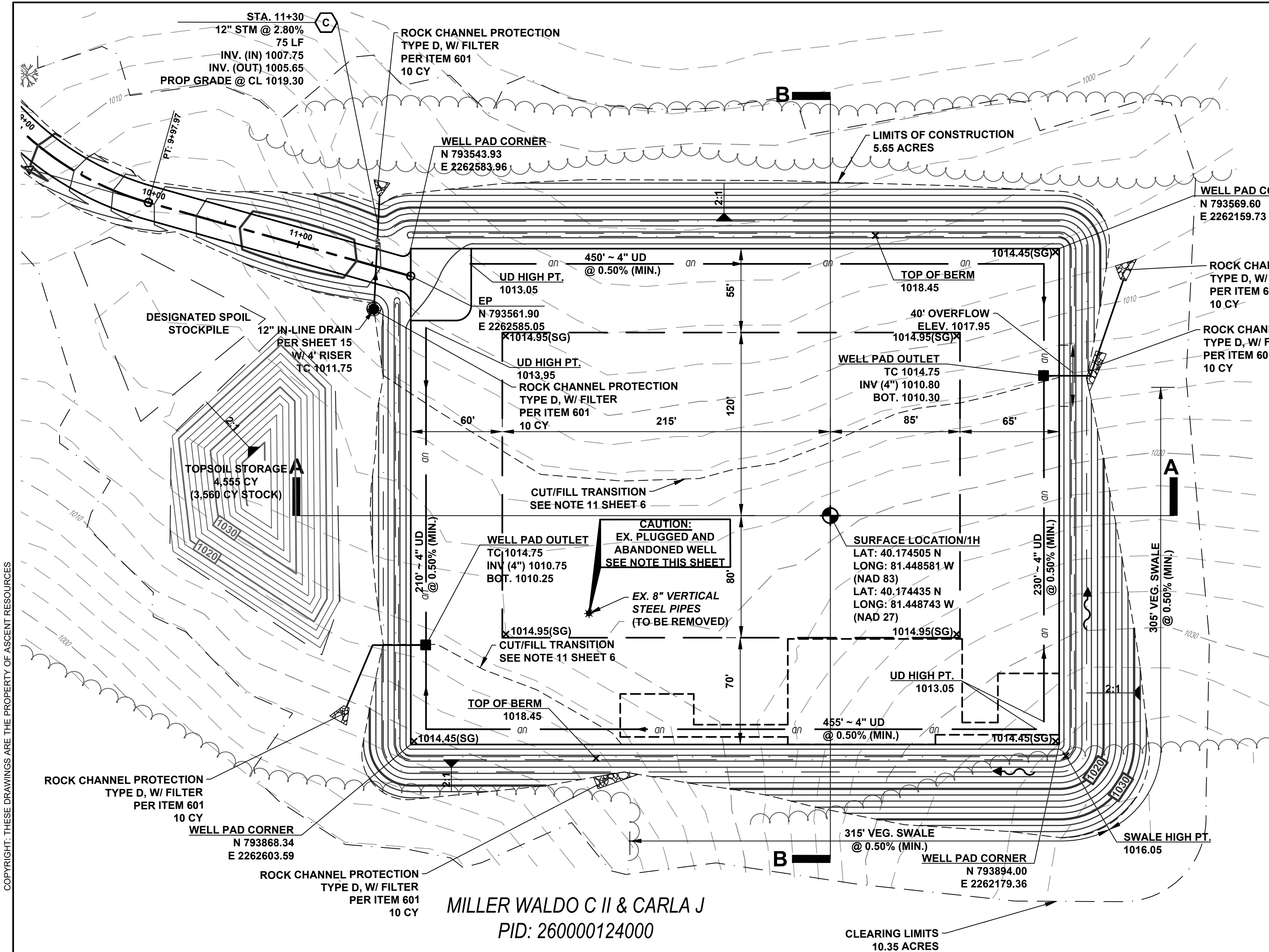
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WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

OVERALL PROPOSED SITE PLAN

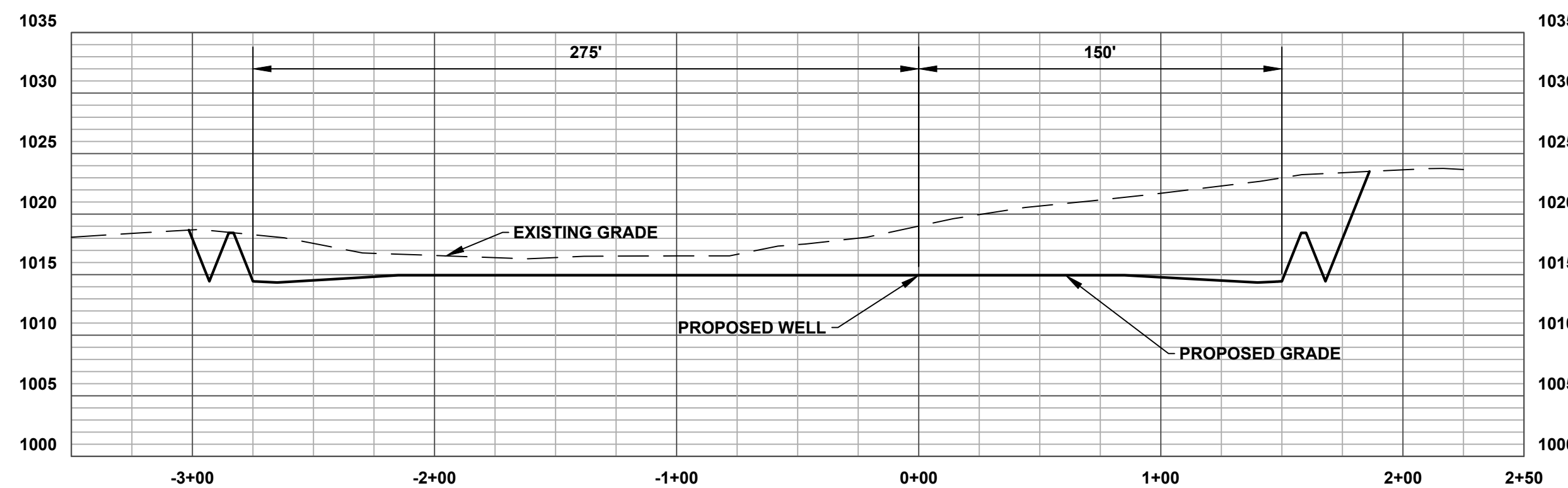
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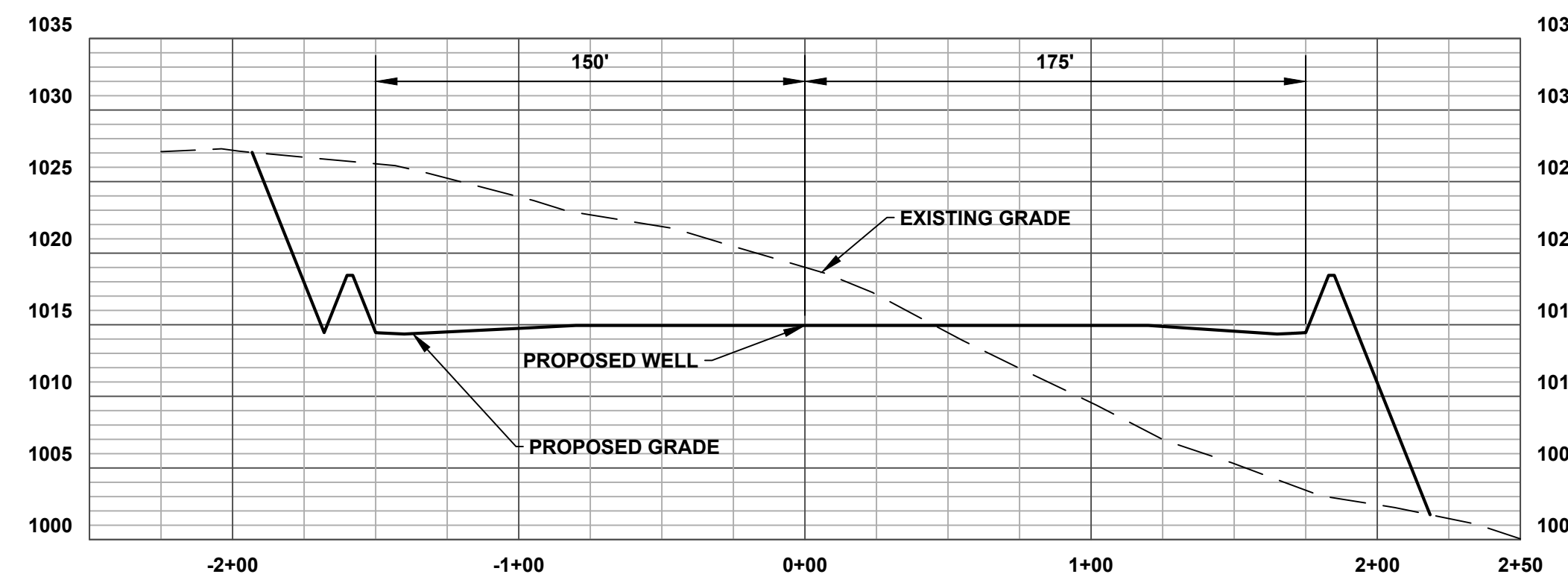
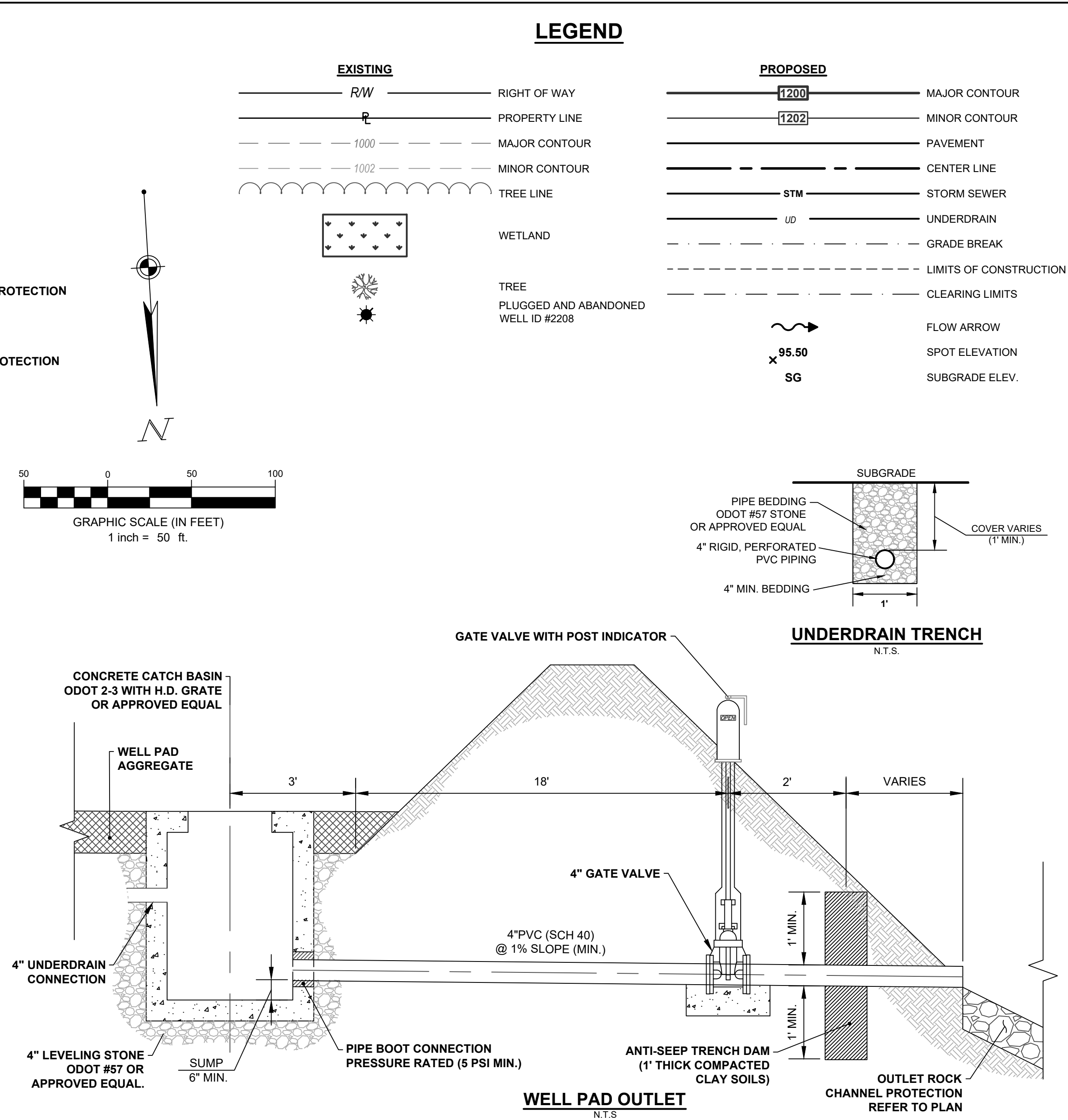
EXISTING PLUGGED AND ABANDONED WELL NOTE:

THE LOCATION OF THE EXISTING PLUGGED AND ABANDONED WELL HAS BEEN NOTED ON THIS PLAN BASED ON OWNER-PROVIDED INFORMATION AND LIMITED FIELD OBSERVATION. CONTRACTOR TO USE EXTREME CAUTION WHILE EXCAVATING IN THE AREA.

THE CONTRACTOR SHALL EXPOSE THE PLUGGED WELL TO AN ELEVATION OF 2-FT BELOW THE PROPOSED WELL PAD SUBGRADE. ASCENT RESOURCES WILL MAKE ARRANGEMENTS TO HAVE THE EXPOSED WELL INSPECTED TO VERIFY IF THE EXISTING PLUG CONFORMS TO OAC 1509.9-11. IF IT IS DETERMINED TO CONFORM, THE PLUG WILL BE CUT AT THIS ELEVATION, AND RE-LABELLED WITH A STEEL PLATE/COVER. IF NOT, ASCENT RESOURCES WILL MAKE ARRANGEMENTS TO PROPERLY PLUG THE WELL, PER OAC 1509.9-11.



A-A SECTION
Scale: 1" = 50' Horiz.; 1" = 10' Vert.



B-B SECTION
Scale: 1" = 50' Horiz.; 1" = 10' Vert.

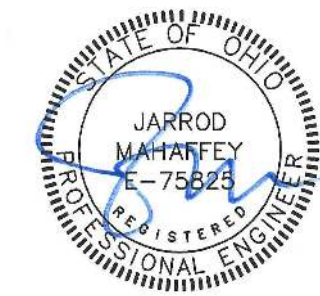
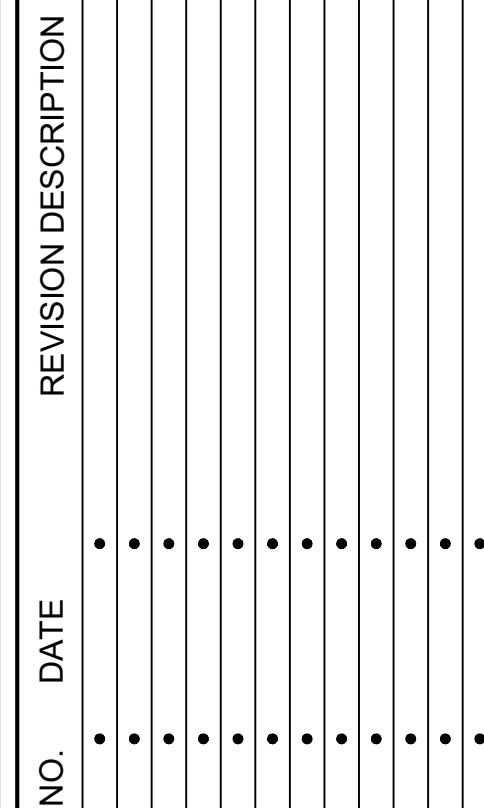
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WALDO MNR GR WELL PAD

MONROE TOWNSHIP GUERNSEY COUNTY, OHIO

SITE PLAN

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ASCENT
RESOURCES

WALDO MNR GR WELL PAD

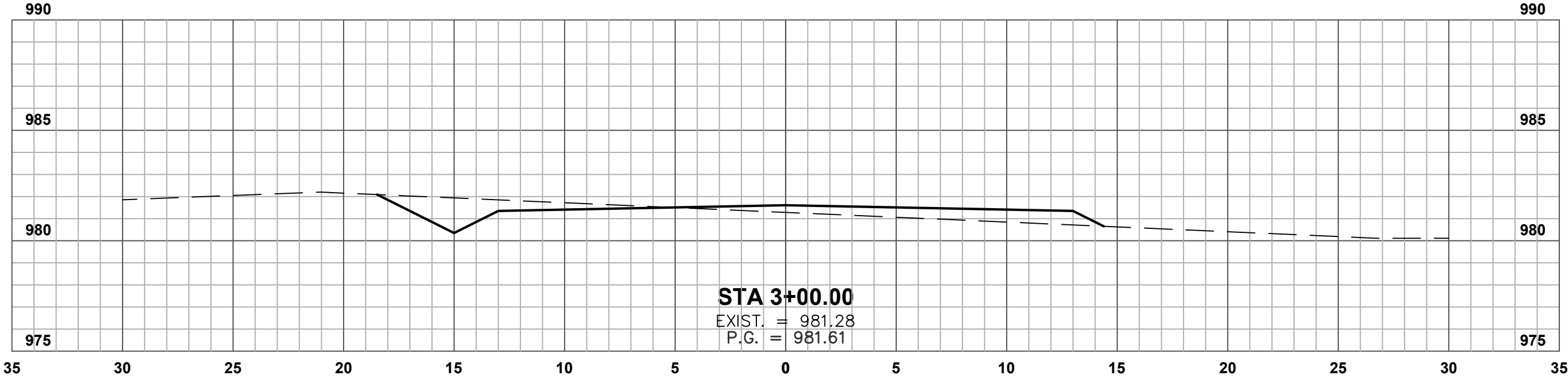
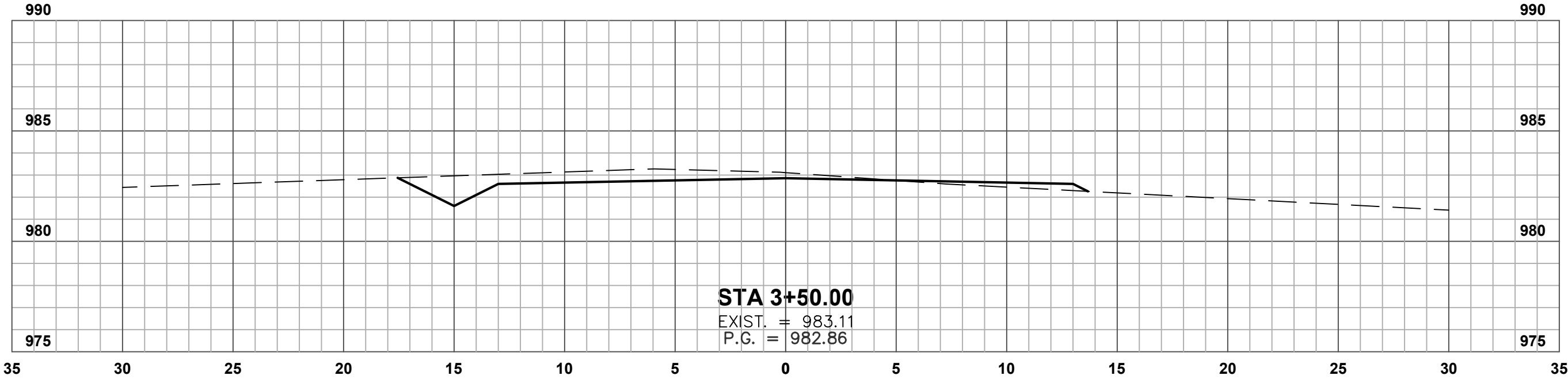
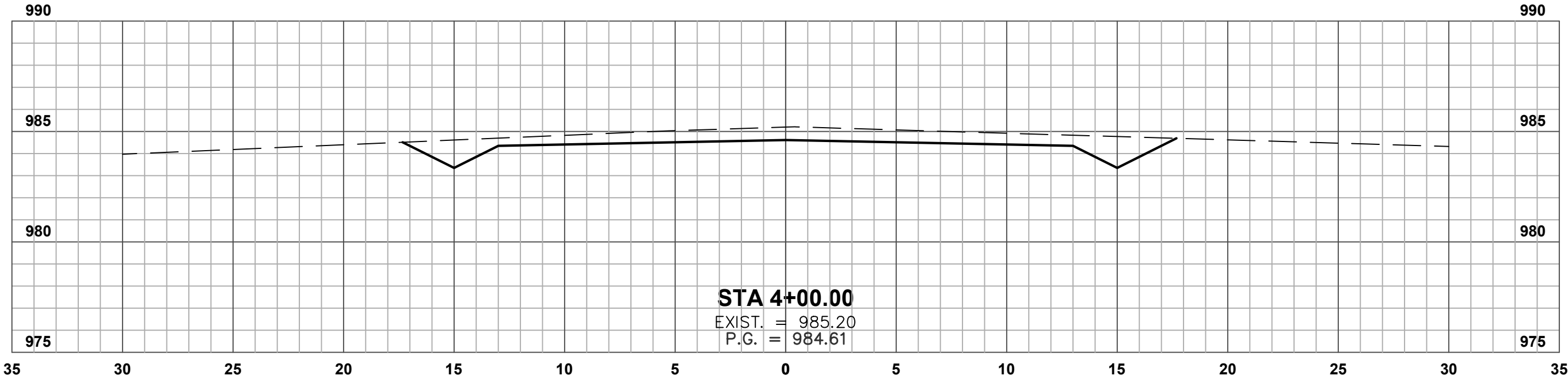
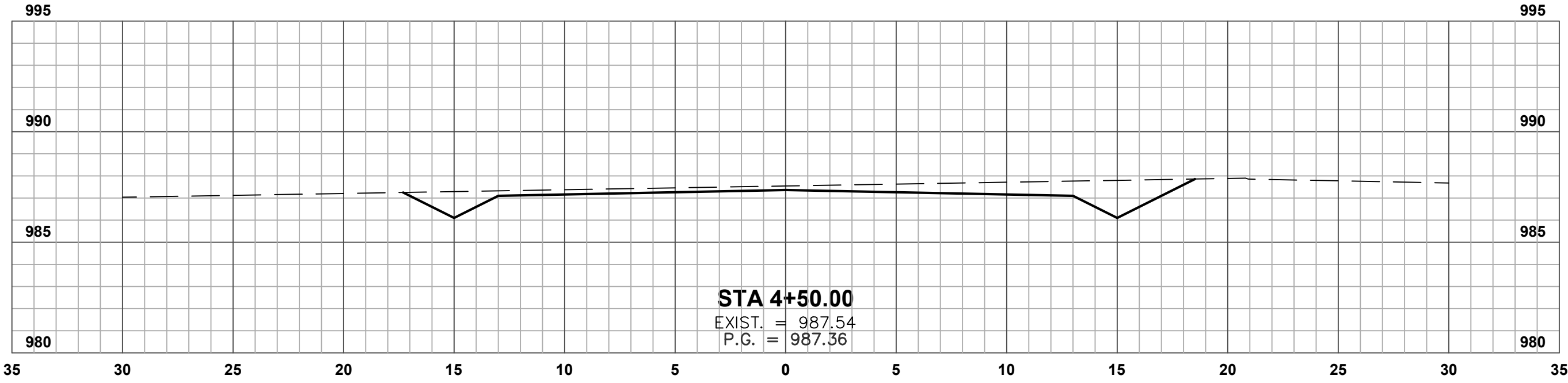
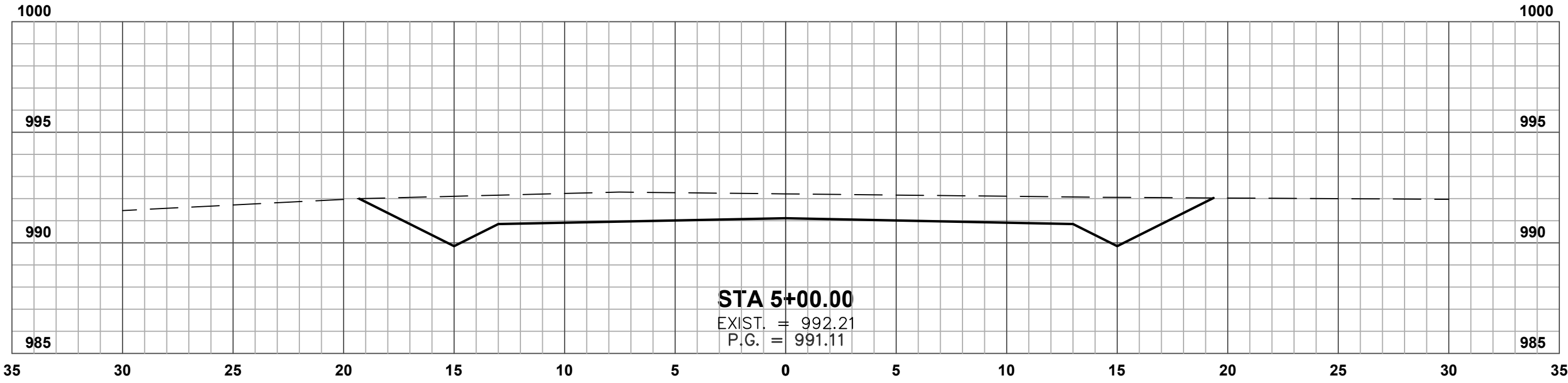
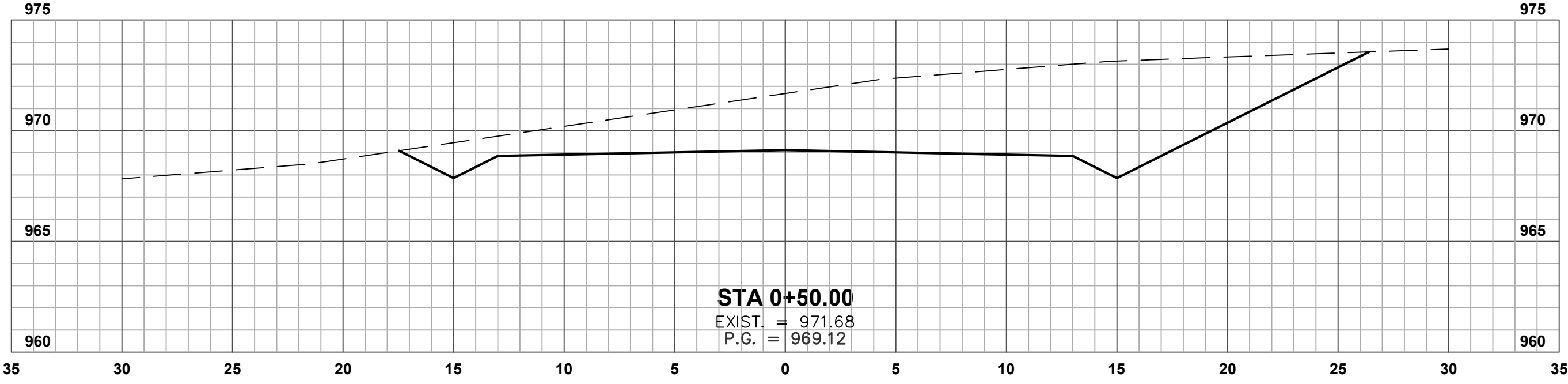
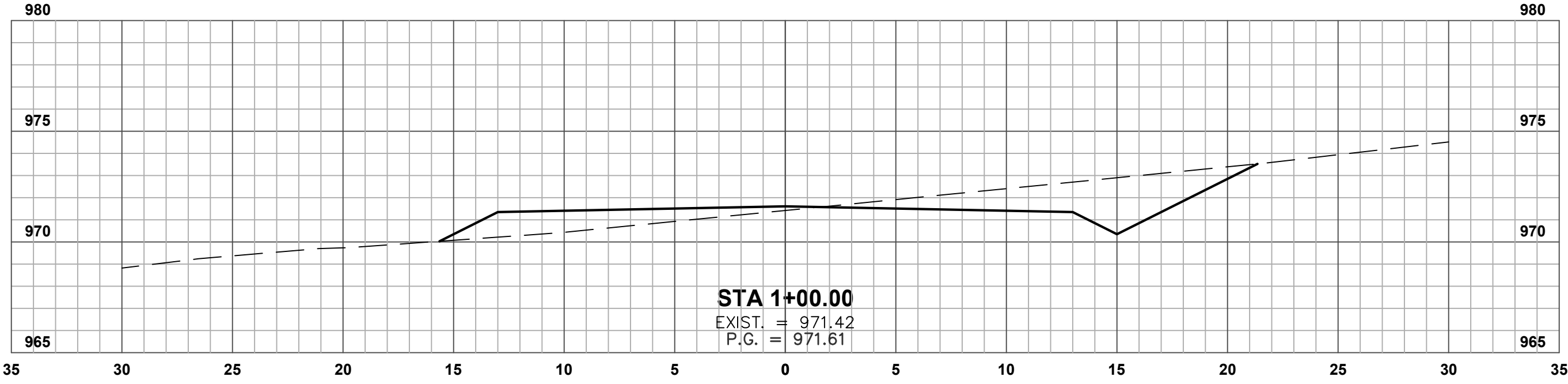
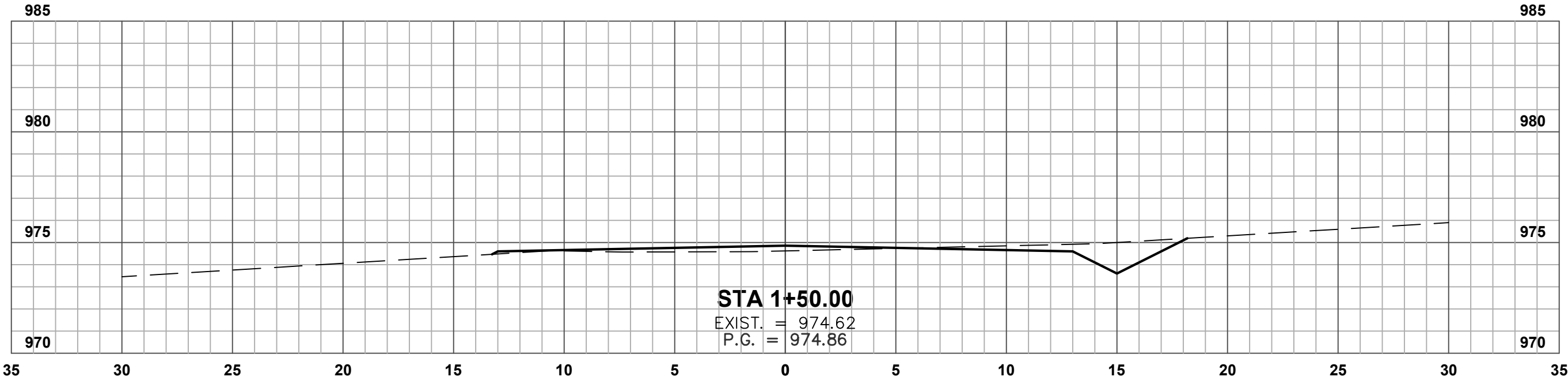
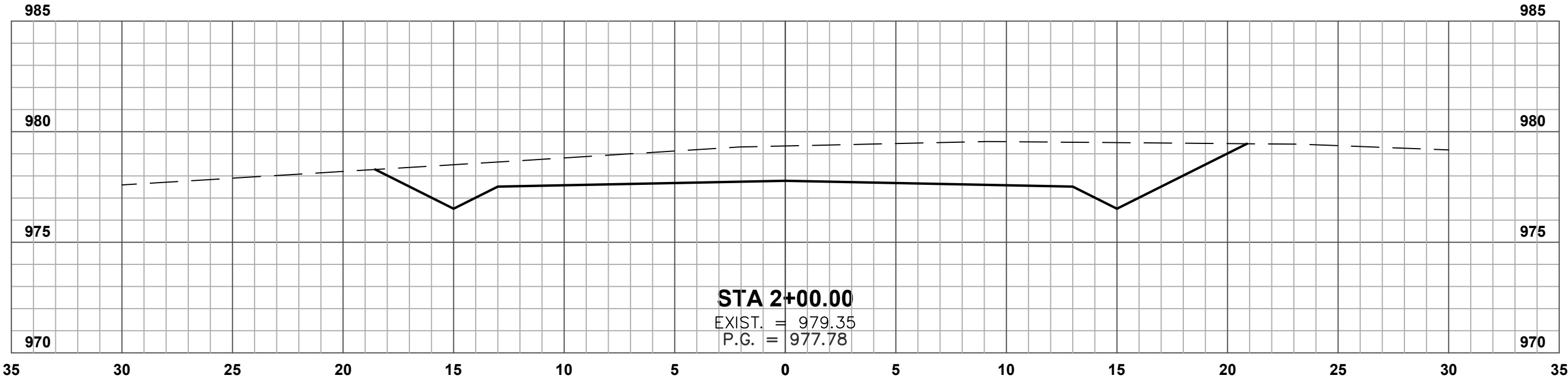
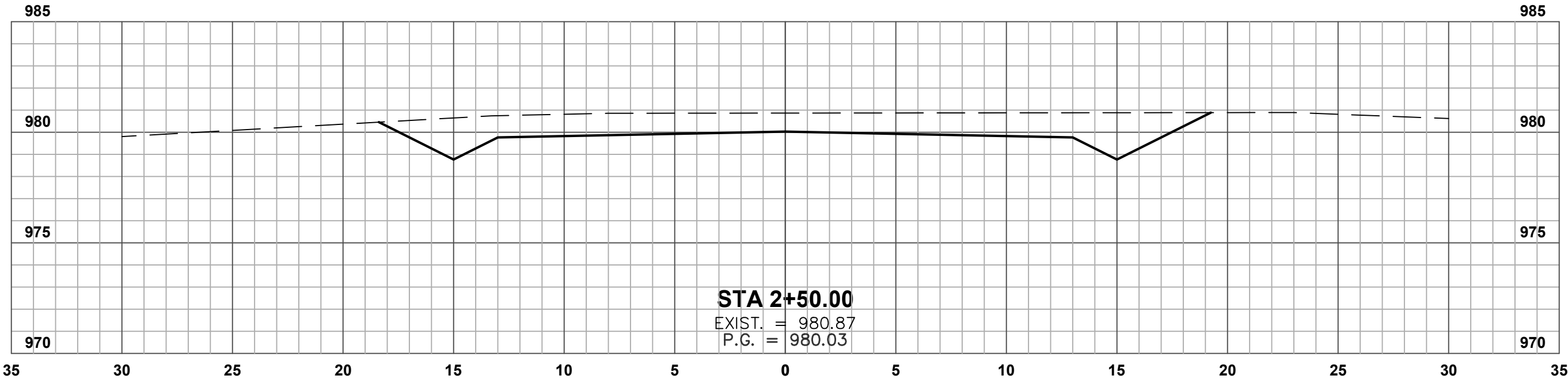
MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

ACCESS DRIVE PLAN & PROFILE

ISSUE:	
CONST. DOCUMENTS	
DATE:	
08.14.2018	
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DESIGN:	JMR
DRAWN:	JMR
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SHEET NO.	

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WALDO MNR GR WELL PAD
MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

ACCESS DRIVE SECTIONS

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DATE:
08.14.2018

JOB NO.: 755215

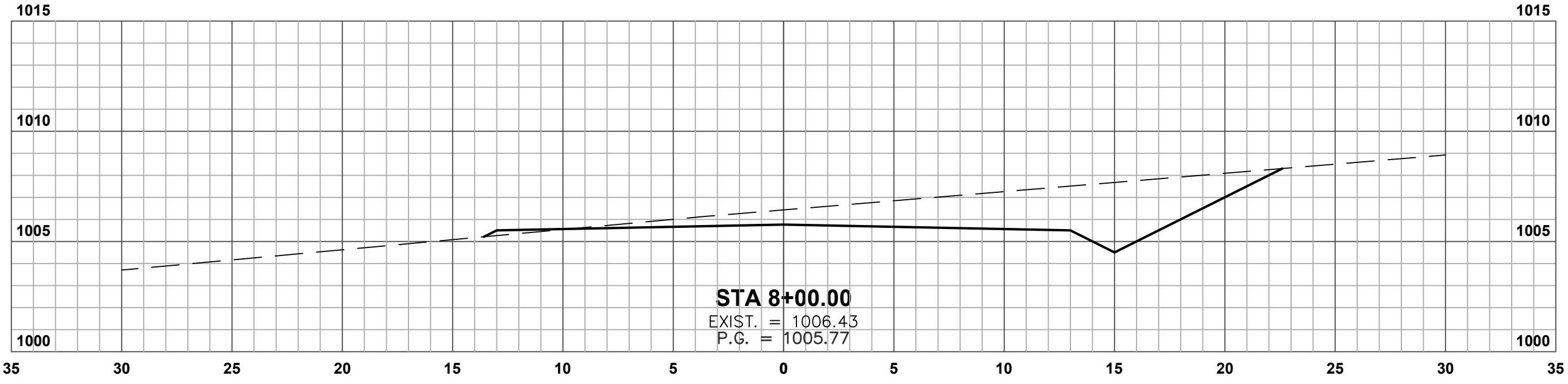
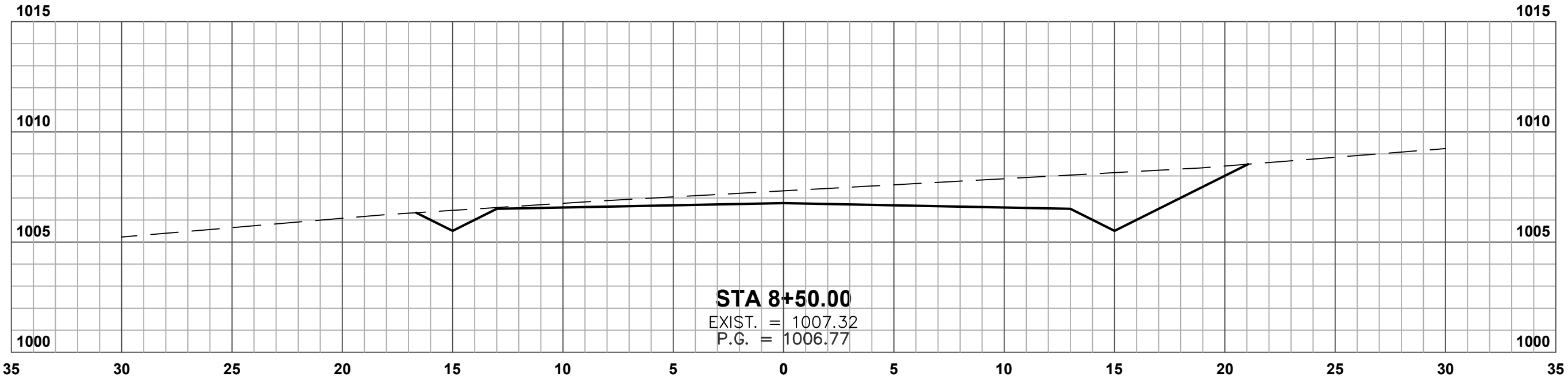
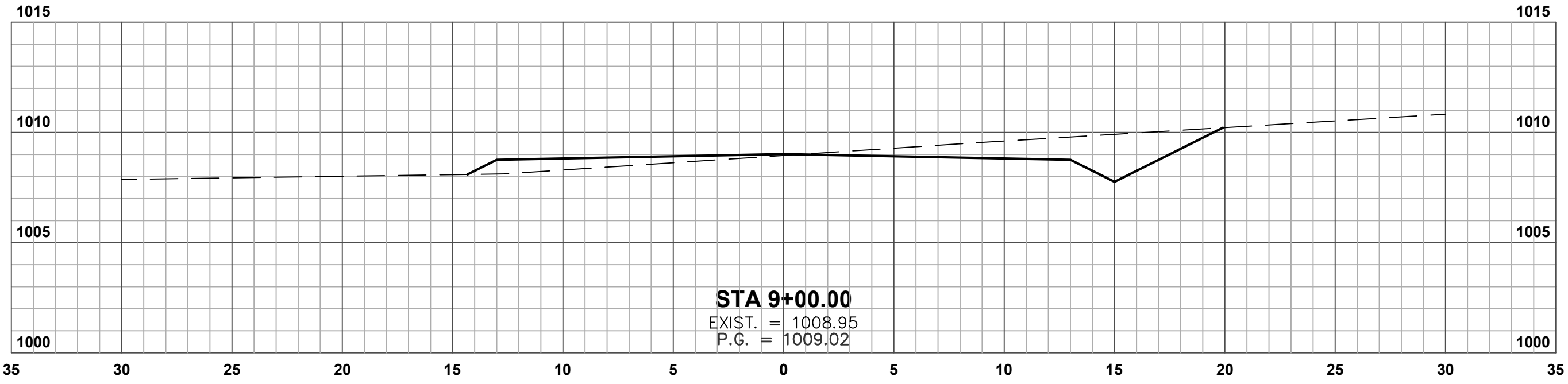
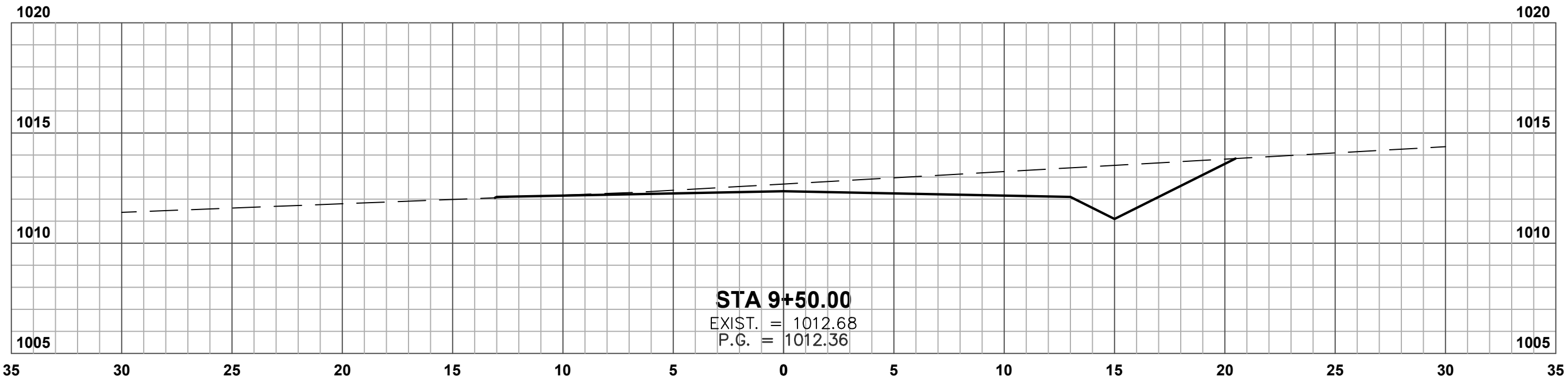
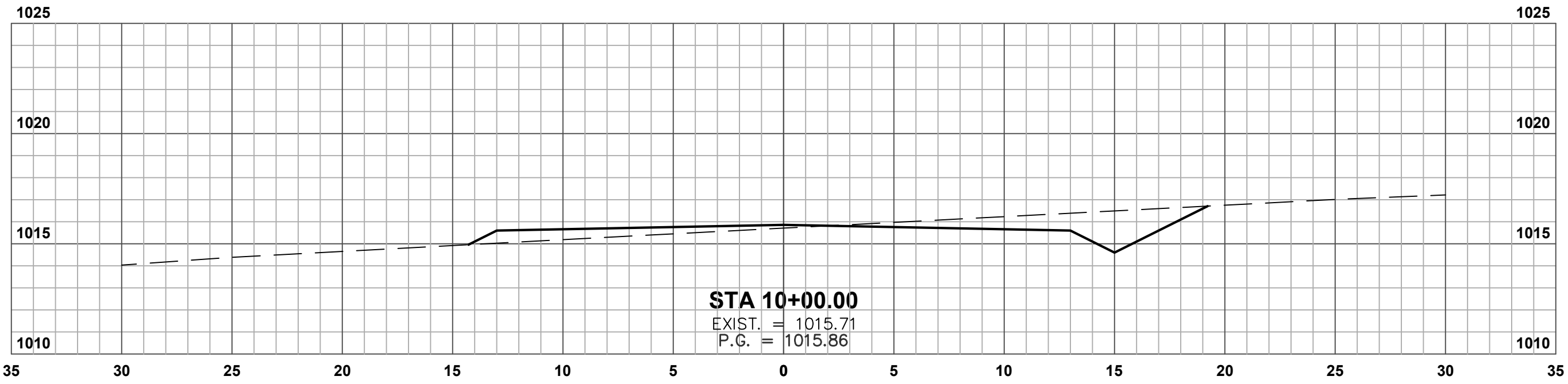
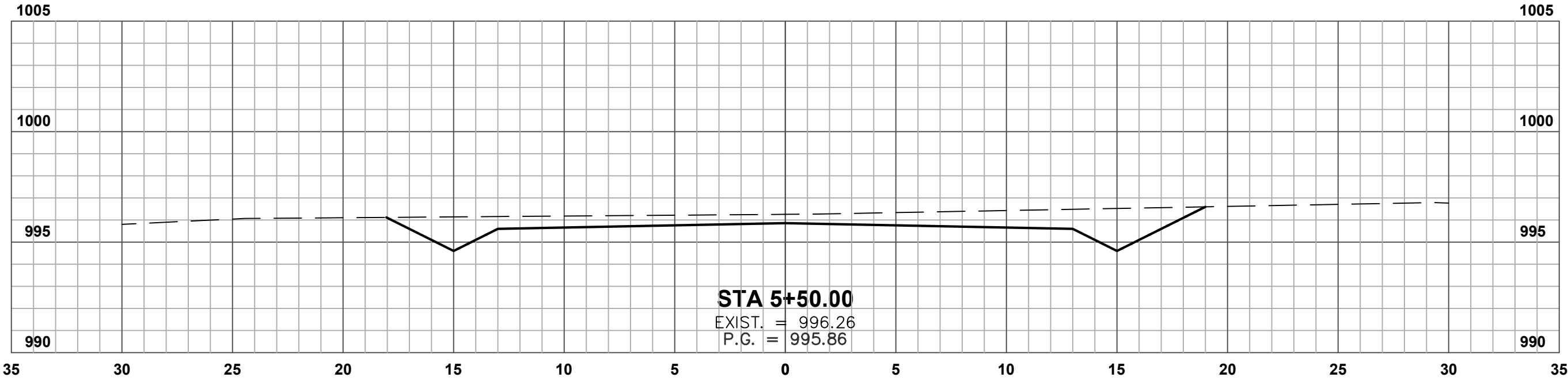
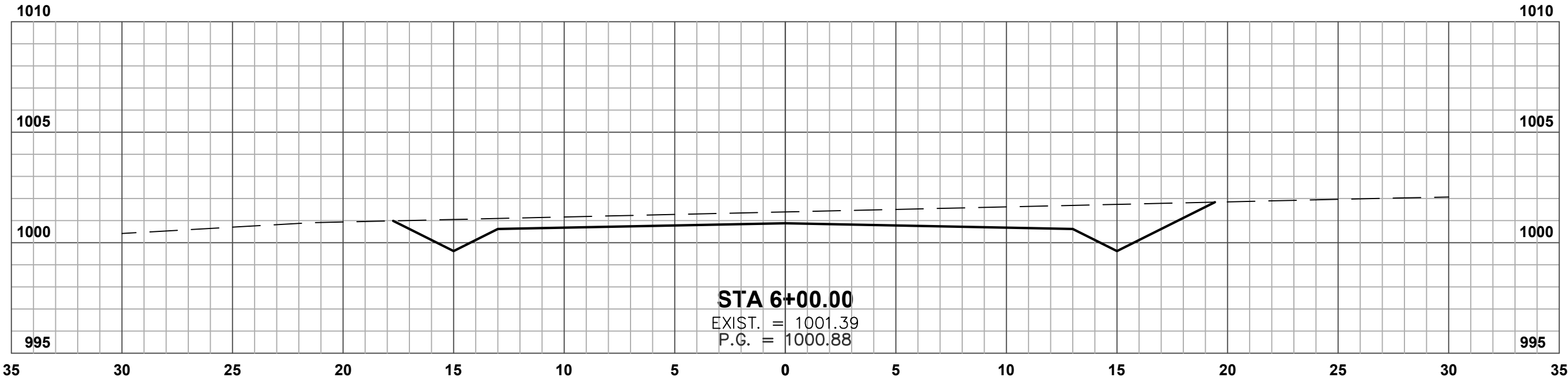
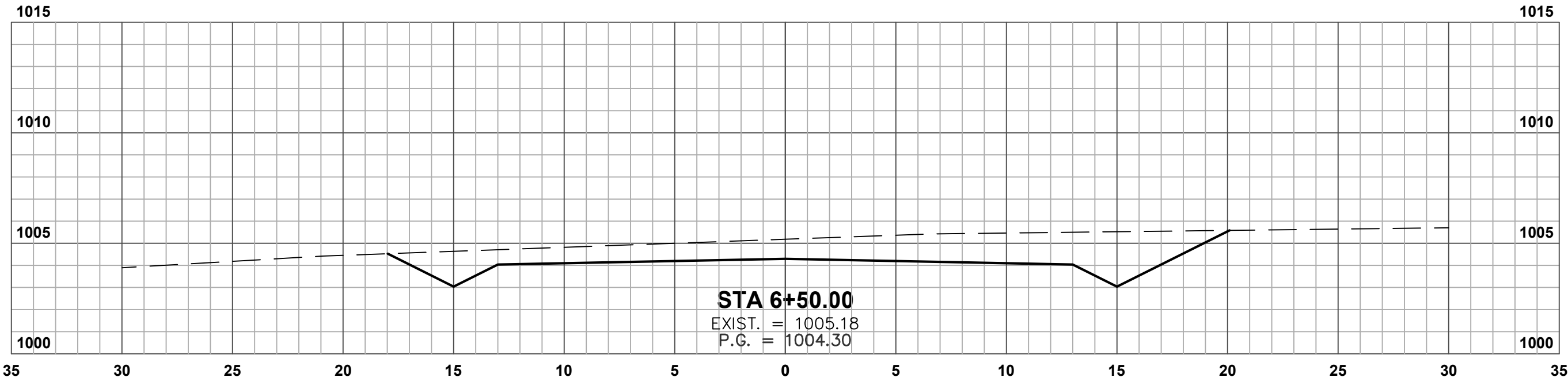
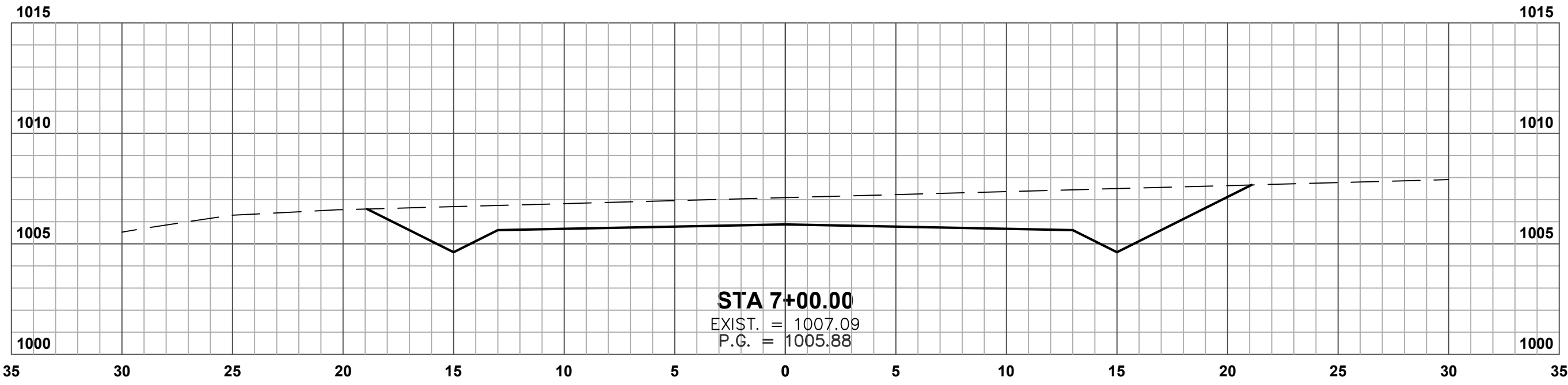
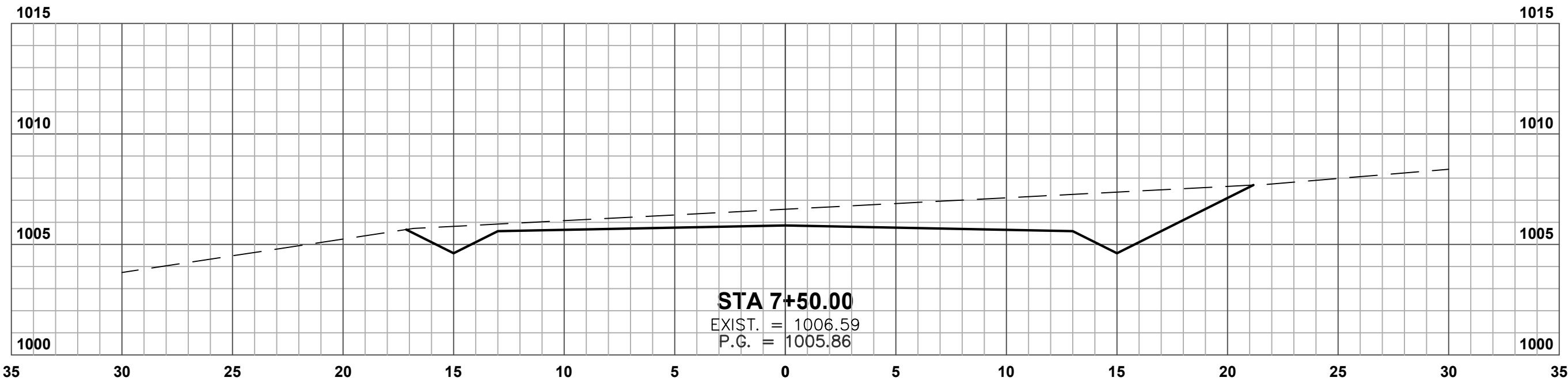
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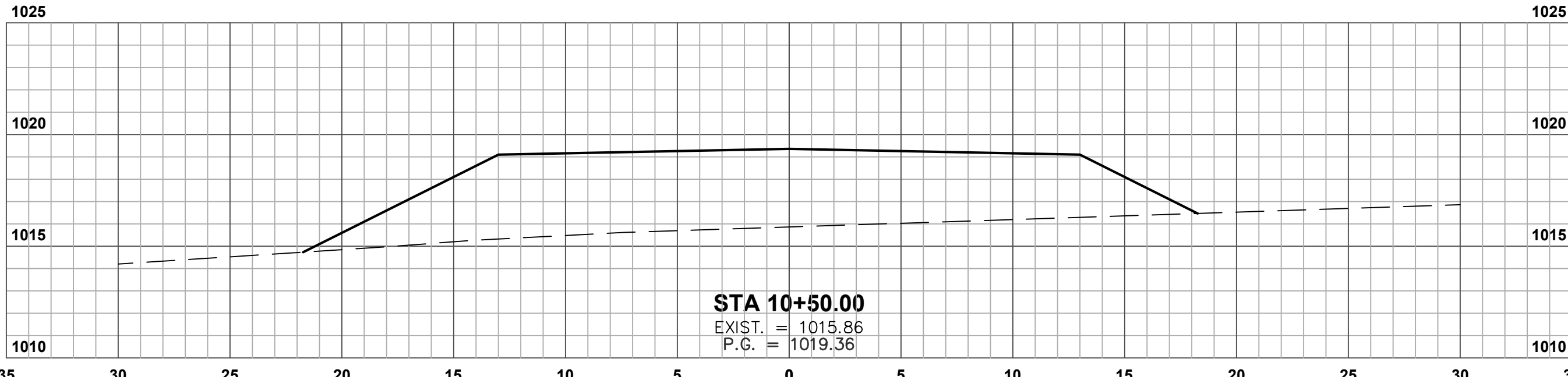
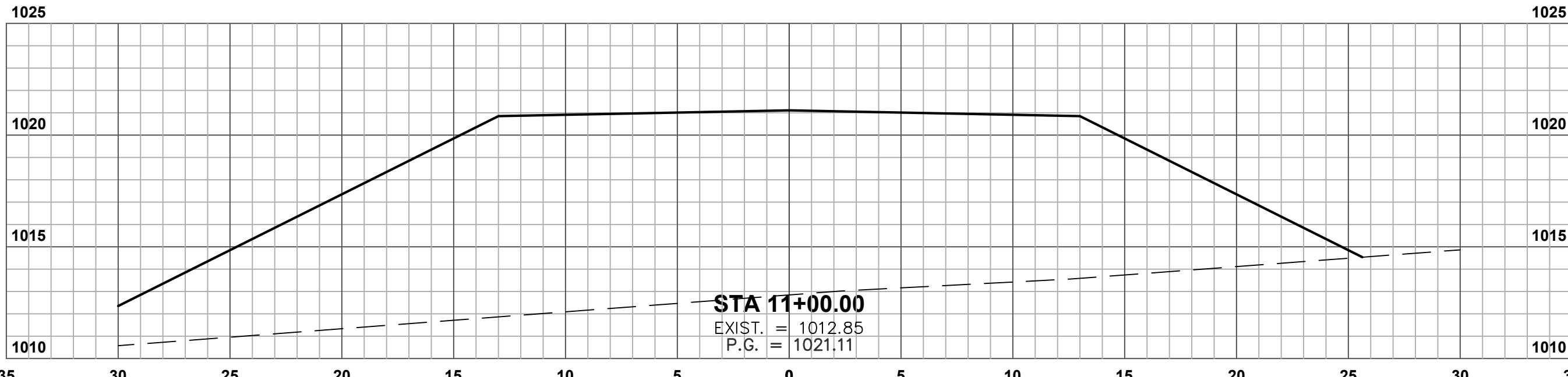
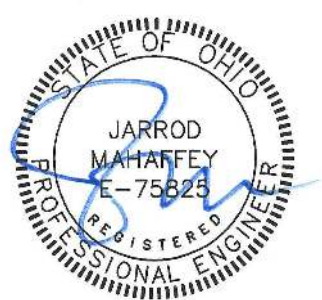
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WALDO MNR GR WELL PAD
MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

ACCESS
DRIVE
SECTIONS

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WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

ACCESS DRIVE SECTIONS

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PROJECT NOTES:

THE PROPOSED PROJECT IS THE CONSTRUCTION OF A NATURAL GAS WELL PAD AND ASSOCIATED ACCESS DRIVE. THIS PROJECT WILL DISTURB 10.35± ACRES.

PRE-CONSTRUCTION RUNOFF CURVE NUMBER - 74
POST-CONSTRUCTION RUNOFF CURVE NUMBER - 74

THE SITE DRAINS TO SOUTHEAST TO UNNAMED TRIBUTARIES OF CLEAR FORK.

THE PROJECT SITE CONTAINS APPROXIMATELY 6" OF TOPSOIL.

SOILS ON SITE CONSIST OF:		
GILPIN SILT LOAM (GdC)	-	10.0%
GILPIN SILT LOAM (GdD)	-	17.9%
GUERNSEY SILT LOAM (GrD2)		6.0%
KEENE SILT LOAM (KeB)		23.1%
WESTMORE SILT LOAM (WkC2)		43.0%

THE PRE-EXISTING CONDITIONS ON SITE ARE UNDEVELOPED MEADOW AND WOODED AREAS.

EROSION & SEDIMENT CONTROL PLAN NOTES

1. ALL EROSION AND SEDIMENTATION CONTROL SHALL BE PERFORMED ACCORDING TO THIS PLAN.
2. CONTRACTOR SHALL IMPLEMENT BEST MANAGEMENT PRACTICES AS STIPULATED IN THE BEST MANAGEMENT PRACTICES FOR OIL AND GAS WELL SITE CONSTRUCTION, ODNR (APRIL 2005). ADDITIONAL BEST MANAGEMENT PRACTICES SHALL BE IMPLEMENTED AS DICTATED BY CONDITIONS AND GRADE CHANGES TO THE SITE AT NO ADDITIONAL COST TO OWNER THROUGHOUT ALL PHASES OF CONSTRUCTION.
3. CONTRACTOR SHALL MINIMIZE CLEARING AND DISTURBANCE TO THE ENVIRONMENT TO THE MAXIMUM EXTENT POSSIBLE OR AS REQUIRED BY THE GENERAL PERMIT.
4. PERIMETER SEDIMENT BARRIERS SHALL BE IMPLEMENTED AS THE FIRST STEP OF GRADING WITHIN SEVEN (7) DAYS FROM THE START OF CLEARING AND GRUBBING, AND SHALL CONTINUE TO FUNCTION UNTIL THE SLOPE DEVELOPMENT AREA IS RESTABILIZED.
5. PERMANENT SOIL STABILIZATION OF DISTURBED AREAS BY MEANS OF VEGETATION, LANDSCAPE TYPE MULCHING, MATTING, SOD, RIP RAP, AND OTHER APPROVED LANDSCAPING TECHNIQUES TO BE APPLIED AS FOLLOWS:
- WITHIN SEVEN (7) DAYS OF ANY AREA THAT WILL BE DORMANT FOR ONE (1) YEAR OR MORE.
- WITHIN TWO (2) DAYS OF ANY AREA WITHIN 50 FEET OF A STREAM AT FINAL GRADE.
- WITHIN SEVEN (7) DAYS FOR ANY OTHER AREA AT FINAL GRADE.
6. TEMPORARY SOIL STABILIZATION OF DISTURBED AREAS BY MEANS OF TEMPORARY VEGETATION, MULCHING, GEOTEXTILES, SOD, PRESERVATION OF EXISTING VEGETATION, AND OTHER APPROVED TECHNIQUES TO BE APPLIED AS FOLLOWS:
- WITHIN TWO (2) DAYS OF ANY AREA WITHIN 50 FEET OF A STREAM NOT AT FINAL GRADE.
- WITHIN SEVEN (7) DAYS OF ANY AREA THAT WILL BE DORMANT FOR MORE THAN TWENTY ONE (21) DAYS, BUT LESS THAN ONE (1) YEAR.
- PRIOR TO THE ONSET OF WINTER WEATHER FOR AREAS THAT WILL BE IDLE OVER WINTER.
7. TEMPORARY SEEDING, MULCHING, AND FERTILIZER SPECIFICATIONS:
SEEDING: ANNUAL RYEGRASS AT 2.02 #/1,000 S.F.
MULCHING: STRAW MATERIAL SHALL BE UNROTTED SMALL GRAIN STRAW APPLIED AT A RATE OF TWO (2) TON/ACRE, OR 80-100 POUNDS PER 1,000 S.F. MULCH MATERIALS SHALL BE RELATIVELY FREE OF ALL KINDS OF WEEDS AND SHALL BE FREE OF PROHIBITIVE NOXIOUS WEEDS. MULCH SHALL BE SPREAD UNIFORMLY BY HAND OR MECHANICAL MEANS. FROM NOVEMBER 01 THRU MARCH 15 INCREASE THE RATE OF STRAW MULCH TO THREE (3) TON/ACRE.
FERTILIZER: APPLY FERTILIZER AT HALF THE RATE OF PERMANENT APPLICATION AND AS PER STATE DOT SPECIFICATIONS. IF PROJECT CONDITIONS PREVENT FERTILIZING THE SOIL, THEN THIS ITEM MAY BE WAIVED.
8. PERMANENT SEEDING SHALL BE IN ACCORDANCE WITH ASCENT RESOURCES, LLC STANDARD SPECIFICATIONS.
9. SLOPES SHALL BE LEFT IN A ROUGHENED CONDITION DURING THE GRADING PHASE TO REDUCE RUNOFF VELOCITIES AND EROSION. ALL SLOPES 3:1 OR GREATER THAN 3:1 SHALL BE FERTILIZED, SEEDED, AND CURLEX BLANKETS BY AMERICAN EXCELSIOR COMPANY, NORTH AMERICAN GREEN, INC. OR AN APPROVED EQUAL AS SPECIFIED IN THE PLANS SHALL BE INSTALLED ON THE SLOPES.
10. NO SOLID (OTHER THAN SEDIMENT) OR LIQUID WASTE, INCLUDING BUILDING MATERIALS, SHALL BE DISCHARGED IN STORM WATER RUNOFF. ALL NON-SEDIMENT POLLUTANTS MUST BE DISPOSED OF IN ACCORDANCE WITH LOCAL, STATE, AND FEDERAL GUIDELINES. WASH OUT OF CEMENT TRUCKS SHOULD OCCUR IN DESIGNATED CONTAINMENT AREAS. WHERE WASHINGS CAN BE REMOVED AND PROPERLY DISPOSED OFF-SITE WHEN THEY HARDEN. STORAGE TANKS SHOULD ALSO BE LOCATED IN CONTAINMENT AREAS. IN ADDITION, SUFFICIENT OIL AND GREASE ABSORBING MATERIALS AND FLOTATION BOOMS TO CLEAN AND CONTAIN FUEL AND CHEMICAL SPILLS MUST BE KEPT ON SITE.
11. RUBBISH, TRASH, GARBAGE, LITTER, OR OTHER SUCH MATERIALS SHALL BE DISPOSED INTO SEALED CONTAINERS. MATERIALS SHALL BE PREVENTED FROM LEAVING THE SITE THROUGH THE ACTION OF WIND OR STORM WATER DISCHARGE INTO DRAINAGE DITCHES OR WATERS OF THE STATE.
12. DUST CONTROL USING ASCENT RESOURCES, LLC APPROVED MATERIAL MUST BE PERFORMED WHEN NECESSARY.
13. ON-SITE AND OFF-SITE STOCKPILE AND BORROW AREAS SHALL BE PROTECTED FROM EROSION AND SEDIMENTATION BY THE USE OF BEST MANAGEMENT PRACTICES. THESE AREAS MUST BE SHOWN IN THE SITE MAP AND PERMITTED IN ACCORDANCE WITH GENERAL PERMIT REQUIREMENTS. AT A MINIMUM SILT FENCE TO BE PLACED AT PERIMETER OF STOCKPILE AREA TO PREVENT SOIL FROM LEAVING THE STOCKPILE AREA.
14. ALL MATERIALS SPILLED, DROPPED, WASHED, OR TRACKED ONTO THE ROADWAYS OR INTO THE STORM SEWERS MUST BE REMOVED IMMEDIATELY.
15. ALL CONSTRUCTION SHALL BE STABILIZED AT THE END OF EACH DAY; THIS INCLUDES BACKFILLING OF TRENCHES FOR UTILITY CONSTRUCTION AND PLACEMENT OF GRAVEL OR ASPHALT FOR ROAD CONSTRUCTION.

EROSION & SEDIMENT CONTROL MAINTENANCE NOTES

ALL CONTROL MEASURES STATED IN THIS PLAN SHALL BE MAINTAINED IN FULLY FUNCTIONAL CONDITION UNTIL TEMPORARY OR PERMANENT STABILIZATION OF THE SITE IS ACHIEVED. ALL EROSION AND SEDIMENTATION CONTROL MEASURES SHALL BE INSPECTED BY A QUALIFIED PERSON IN ACCORDANCE TO THE CONTRACT DOCUMENTS OR THE APPLICABLE PERMIT, WHICHEVER IS MORE STRINGENT, AND REPAIRED ACCORDING TO THE FOLLOWING:

1. ALL SEEDED AREAS SHALL BE CHECKED REGULARLY TO ENSURE THAT A GOOD STANDING OF GRASS IS MAINTAINED. AREAS SHOULD BE FERTILIZED, WATERED, AND RESEDED AS NEEDED.
2. SILT FENCES, AND CHECK DAMS SHALL BE REPAIRED TO THEIR ORIGINAL CONDITION IF DAMAGED. SEDIMENT ACCUMULATION MUST BE REMOVED WHEN SEDIMENT HEIGHT REACHES ONE-HALF THE HEIGHT OF THE SILT FENCE OR CHECK DAM.
3. SEDIMENTATION TRAPS SHALL BE MAINTAINED IN OPERATIONAL CONDITIONS AT ALL TIMES. SEDIMENT MUST BE REMOVED FROM TRAPS WHEN THE DESIGN CAPACITY HAS BEEN REDUCED BY 40%.
4. MINIMIZE OFF-SITE SEDIMENT TRACKING OF VEHICLES BY THE USE OF STONE MATERIAL IN ALL CONSTRUCTION ENTRANCES, ALONG WITH REGULARLY SCHEDULED SWEEPING/GOOD HOUSEKEEPING. STABILIZED CONSTRUCTION ENTRANCES TO BE PROPERLY MAINTAINED BY GENERAL CONTRACTOR AND IN GOOD WORKING ORDER AT ALL TIMES; THIS MAY REQUIRE PERIODIC TOP DRESSING OF THE STONE AS CONDITIONS DEMAND.
5. THE TEMPORARY PARKING AND STORAGE AREA SHALL BE KEPT IN GOOD CONDITION (SUITABLE FOR PARKING AND STORAGE) BY GENERAL CONTRACTOR. THIS MAY REQUIRE PERIODIC TOP DRESSING OF THE TEMPORARY PARKING AS CONDITIONS DEMAND.
6. CONTRACTORS AND SUBCONTRACTORS WILL BE RESPONSIBLE FOR REMOVING ALL SEDIMENT FROM THE SITE, INCLUDING SEDIMENT TRAPS AND STORM SEWER PIPES. SEDIMENT DEPOSITION DURING SITE STABILIZATION MUST ALSO BE REMOVED.
7. ALL ROCK CHANNEL PROTECTION MUST BE PLACED OVER GEOTEXTILE FILTER.
8. ROCK CHECK DAMS SHOULD BE ROUTINELY CLEANED ONCE SEDIMENT BEGINS TO APPEAR ON THE UPSTREAM SIDE OF THE ROCK.
9. CONTAINERS SHALL BE AVAILABLE FOR DISPOSAL OF DEBRIS, TRASH, HAZARDOUS OR PETROLIEUM WASTES. ALL CONTAINERS MUST BE COVERED AND LEAK-PROOF. ALL WASTE MATERIAL SHALL BE DISPOSED OF AT FACILITIES APPROVED FOR THE PERTINENT MATERIAL
10. AREA SHALL BE DESIGNATED BY CONTRACTOR AND SHOWN ON SWPPP MAP FOR MIXING OR STORAGE OF COMPOUNDS SUCH AS FERTILZERS, LIME ASPHALT, OR CONCRETE, THESE DESIGNATED AREAS SHALL BE LOCATED AWAY FROM WATERCOURSES, DRAINAGE DITCHES, FIELD DRAINS, OR OTHER STORMWATER DRAINAGE AREA.
11. EQUIPMENT FUELING & MAINTENANCE SHALL BE IN DESIGNATED AREAS ONLY.
12. ALL CONTAMINATED SOIL MUST BE TREATED AND/OR DISPOSED IN AN OHIO EPA APPROVED SOLID WASTE MANAGEMENT FACILITY OR HAZARDOUS WASTE TREATMENT, STORAGE OR DISPOSAL FACILITIES.
13. THE CONTRACTOR SHALL CONTACT ASCENT RESOURCES, LLC IN THE EVENT OF ANY SPILL OR THE PRESENCE OF SHEEN.



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WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

EROSION & SEDIMENT CONTROL NOTES

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DATE:
08.14.2018

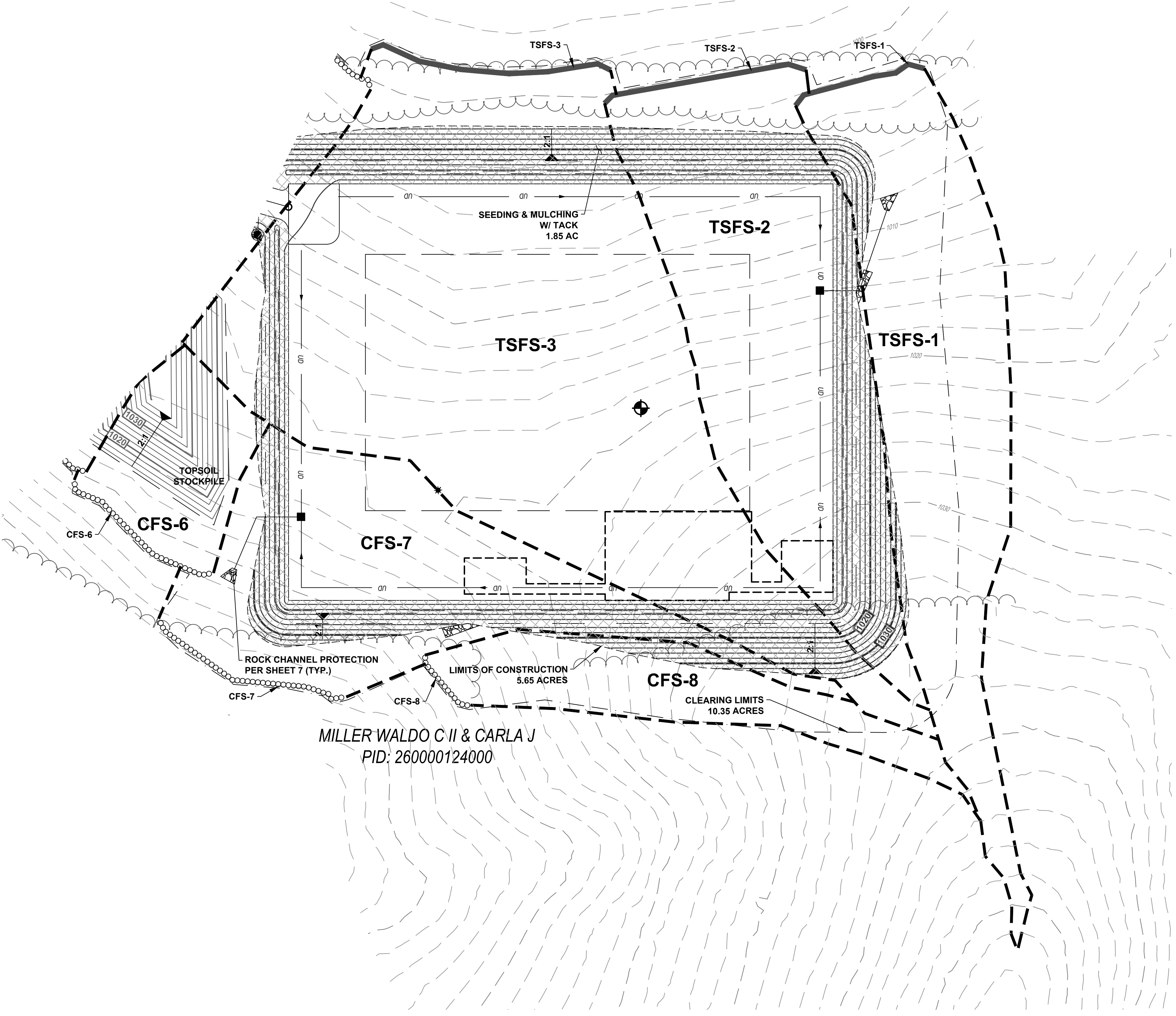
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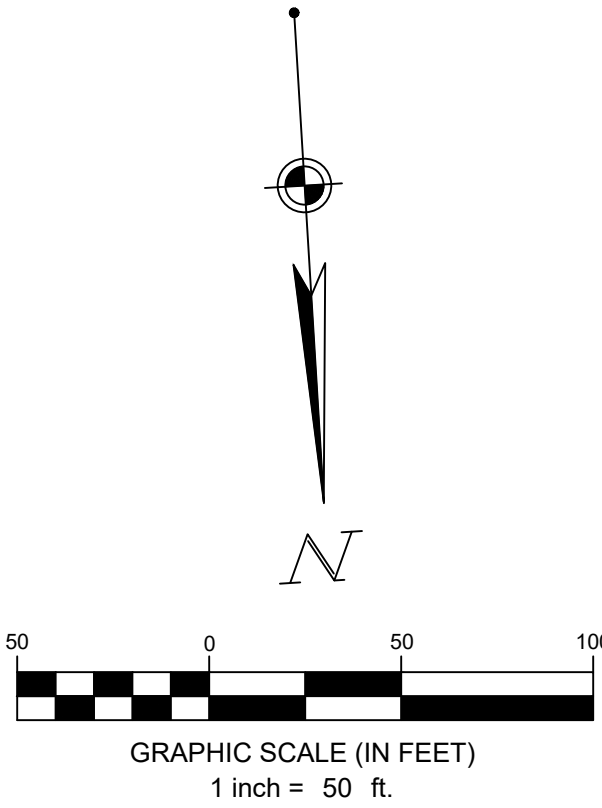
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SHEET NO.
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LEGEND

EXISTING		PROPOSED	
	PROPERTY LINE		MAJOR CONTOUR
	MAJOR CONTOUR		MINOR CONTOUR
	MINOR CONTOUR		PAVEMENT
	TREE LINE		CENTER LINE
	TREE		STORM SEWER
	PLUGGED AND ABANDONED WELL ID #2208		UNDERDRAIN
			GRADE BREAK
			LIMITS OF CONSTRUCTION
			CLEARING LIMITS
			FILTER SOCK, PER SHEET 15
			TRIPLE STACK FILTER SOCK, PER SHEET 15
			DRAINAGE AREA BOUNDARY
			SEEDING & MULCHING W/ TACK



COMPOST FILTER SOCK SIZING TABLE

SOCK NO.	DRAINAGE AREA (ACRES)	% SLOPE	SLOPE LENGTH (FT)	SOCK LENGTH (FT)	SOCK SIZE (INCHES)
CFS-6	0.35	<10	145	140	18
CFS-7	0.25	<10	160	95	18
CFS-8	1.05	<10	435	155	32
CFS-9	0.45	<10	460	60	32
TSFS-1	1.20	<10	715	110	TS
TSFS-2	1.35	<10	525	165	TS
TSFS-3	2.80	<10	635	195	TS

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WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

EROSION & SEDIMENT CONTROL PLAN

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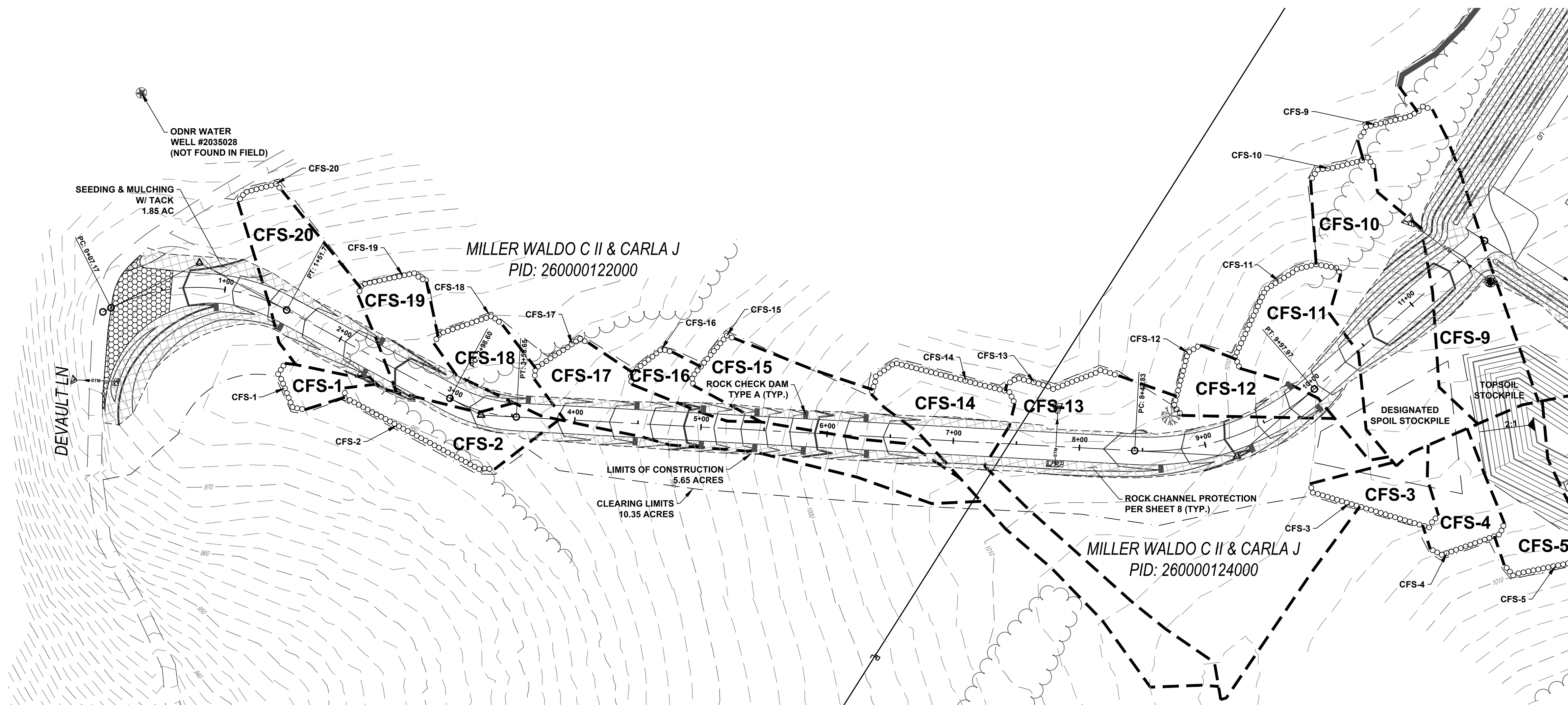
WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

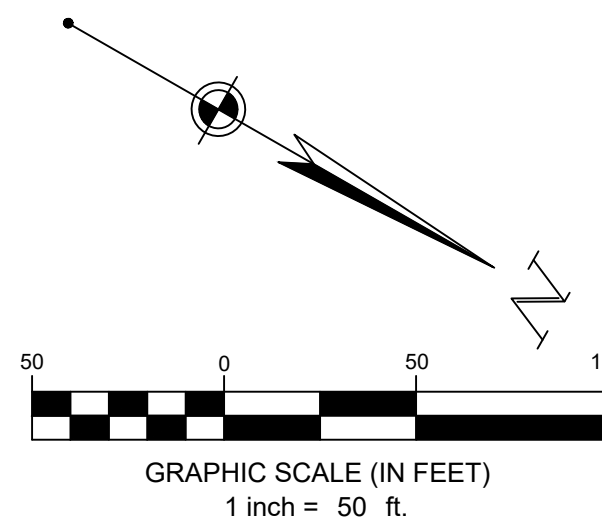
EROSION & SEDIMENT CONTROL PLAN

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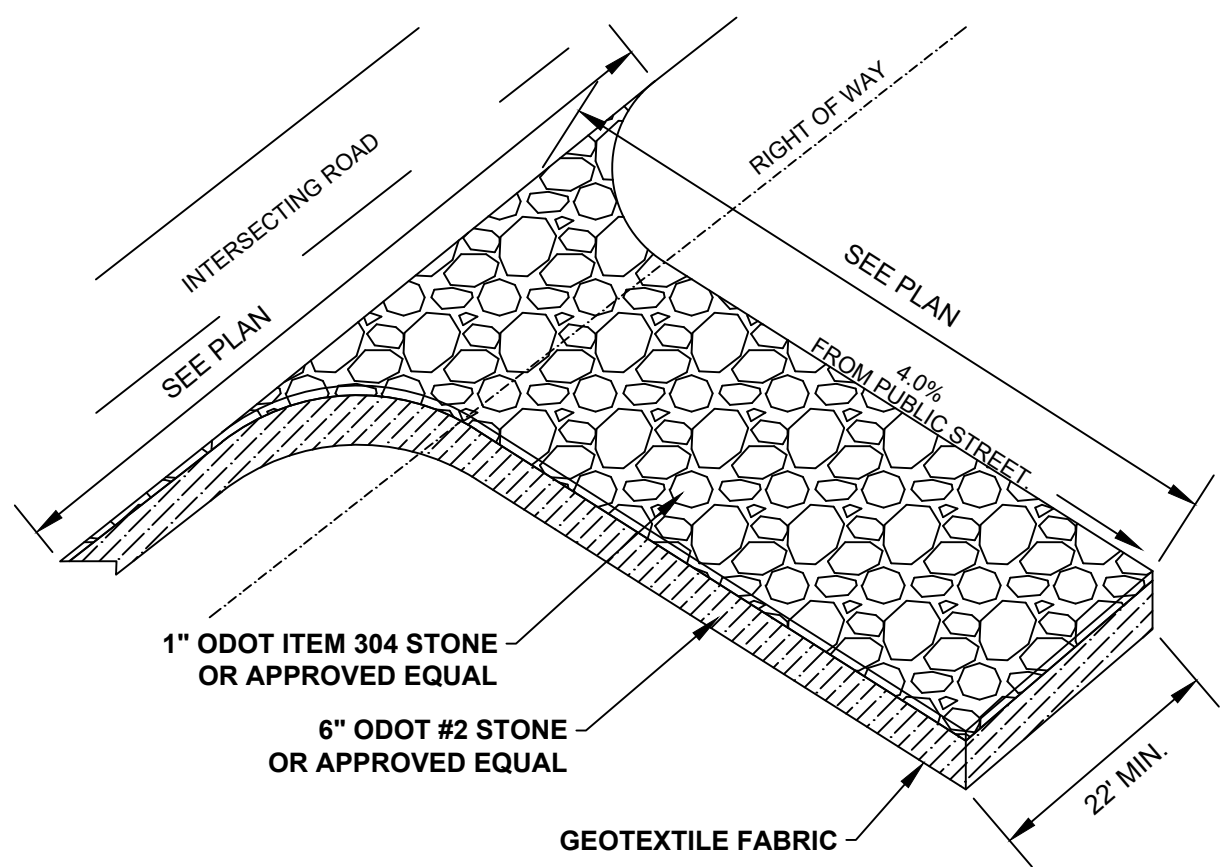
COMPOST FILTER SOCK SIZING TABLE					
SOCK NO.	DRAINAGE AREA (ACRES)	% SLOPE	SLOPE LENGTH (FT)	SOCK LENGTH (FT)	SOCK SIZE (INCHES)
CFS-1	0.10	<5	70	50	18
CFS-2	0.15	<10	75	140	18
CFS-3	0.10	<5	60	115	18
CFS-4	0.10	<5	90	70	18
CFS-5	0.20	10	130	80	18
CFS-9	0.35	<10	250	70	18
CFS-10	0.40	<10	240	60	18
CFS-11	0.20	<5	150	140	18
CFS-12	0.10	<10	130	65	18
CFS-13	1.00	<10	300	95	18
CFS-14	0.30	<5	375	140	18
CFS-15	0.20	<10	220	220	18
CFS-16	0.10	<10	105	105	18
CFS-17	0.30	<10	365	365	24
CFS-18	0.10	<5	105	105	18
CFS-19	0.15	<5	155	155	18
CFS-20	0.25	<10	190	190	18



LEGEND

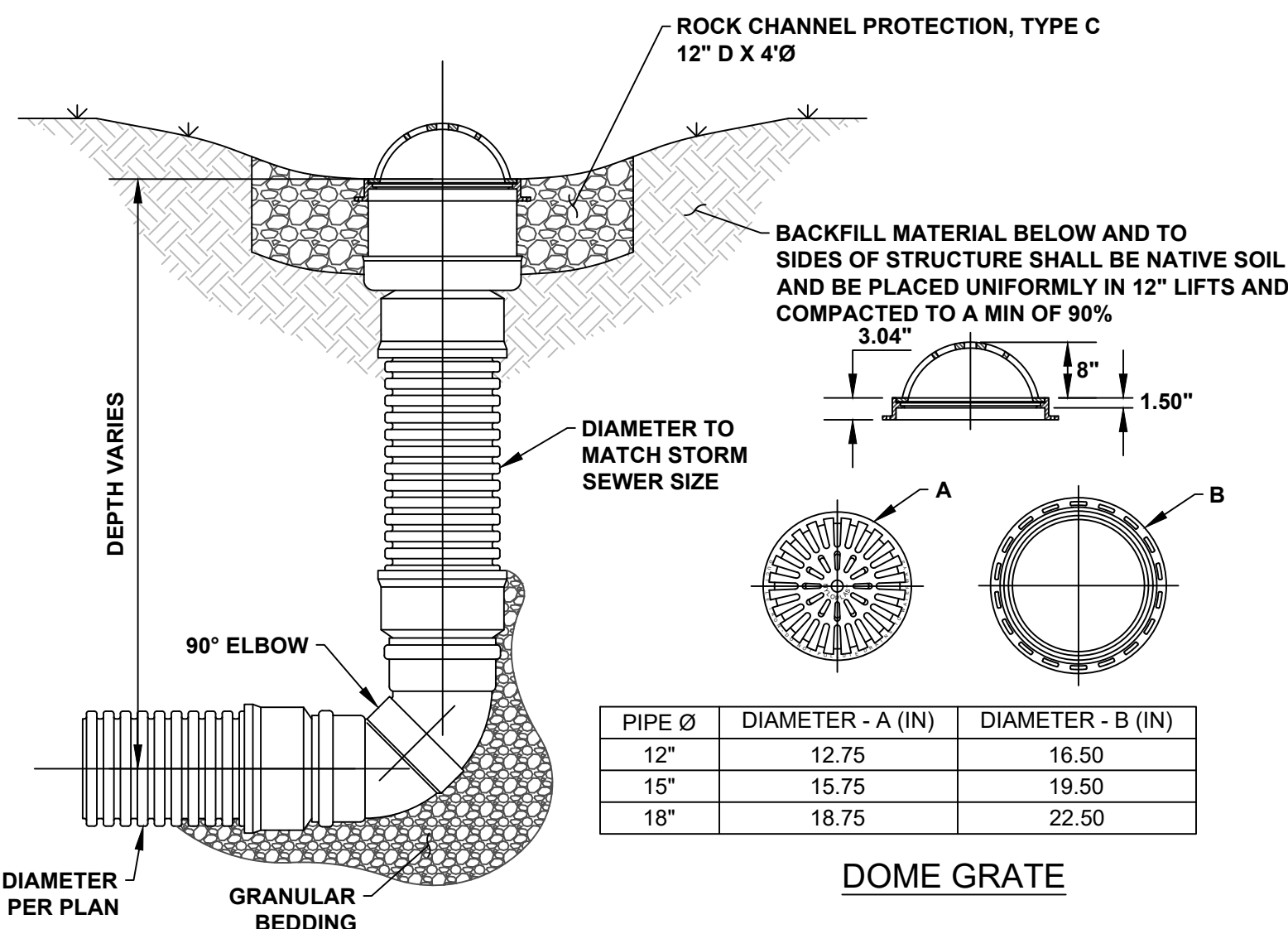
EXISTING		PROPOSED	
	RIGHT OF WAY		MAJOR CONTOUR
	PROPERTY LINE		MINOR CONTOUR
	MAJOR CONTOUR		PAVEMENT
	MINOR CONTOUR		CENTER LINE
	PAVEMENT		STORM SEWER
	TREE LINE		UNDERDRAIN
	TREE		GRADE BREAK
	WATER WELL		LIMITS OF CONSTRUCTION
			CLEARING LIMITS
			FILTER SOCK, PER SHEET 15
			TRIPLE STACK FILTER SOCK, PER SHEET 15
			DRAINAGE AREA BOUNDARY
			ROCK CHECK DAM, PER SHEET 15
			STABILIZED CONSTRUCTION ENTRANCE, PER SHEET 15
			SEEDING & MULCHING W/ TACK

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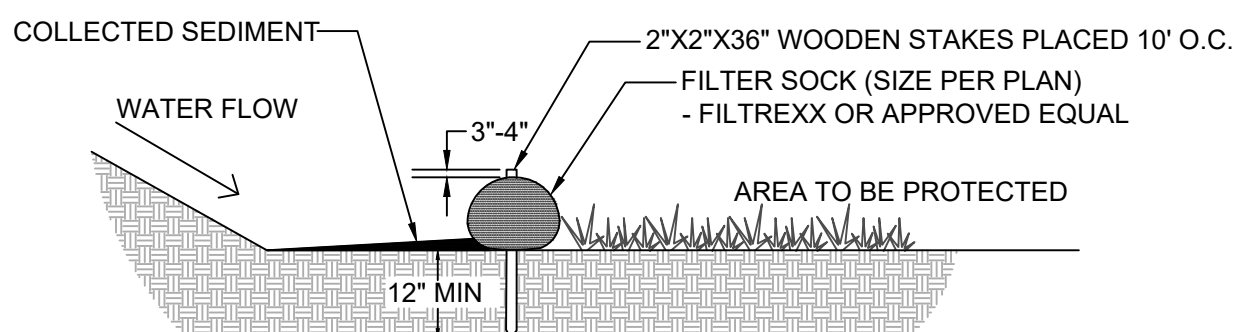
STABILIZED CONSTRUCTION ENTRANCE

N.T.S.



IN-LINE DRAIN W/ DOME GRATE

N.T.S.



SECTION

N.T.S.

MATERIAL NOTES:

1. COMPOST USED FOR FILTER SOCKS SHALL BE WEED, PATHOGEN AND INSECT FREE AND FREE OF ANY REFUSE, CONTAMINANTS OR OTHER MATERIALS TOXIC TO PLANT GROWTH. THEY SHALL BE DERIVED FROM A WELL-DECOMPOSED SOURCE OF ORGANIC MATTER AND CONSIST OF PARTICLES RANGING FROM 3/8" TO 2".

2. FILTER SOCKS SHALL BE 3 OR 5 MIL CONTINUOUS, TABULAR, HDPE 3/8" KNITTED MESH NETTING MATERIAL FILLED WITH COMPOST PASSING THE ABOVE SPECIFICATIONS FOR COMPOST PRODUCTS.

INSTALLATION NOTES:

3. FILTER SOCKS WILL BE PLACED ON A LEVEL LINE ACROSS SLOPES.

4. FILTER SOCKS INTENDED TO BE PERMANENT SHALL BE SEED AT TIME OF INSTALLATION.

5. FILTER SOCKS ARE NOT TO BE USED IN CONCENTRATED FLOW SITUATIONS OR IN RUNOFF CHANNELS.

MAINTENANCE NOTES:

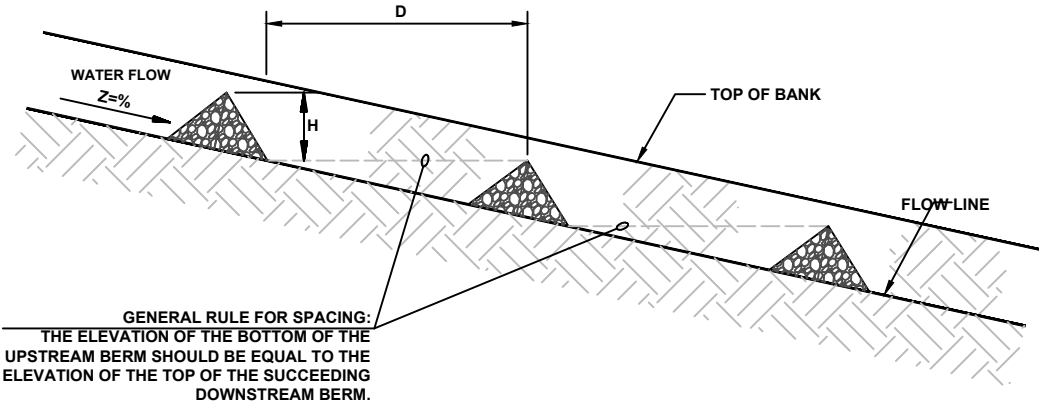
6. ROUTINELY INSPECT FILTER SOCKS AFTER SIGNIFICANT RAIN , KEEPING THEM FUNCTIONAL AT ALL TIMES.

7. REMOVE SEDIMENTS COLLECTED AT THE BASE OF THE FILTER SOCKS WHEN SEDIMENT REACHES 1/3 OF THE EXPOSED HEIGHT OF THE PRACTICE.

8. WHERE THE FILTER SOCK DETERIORATES OR FAILS, IT WILL BE REPAIRED OR REPLACED WITH A MORE EFFECTIVE ALTERNATIVE.

9. FILTER SOCKS WILL BE DISPERSED ON SITE WHEN NO LONGER REQUIRED IN SUCH A WAY AS TO FACILITATE AND NOT OBSTRUCT SEEDING.

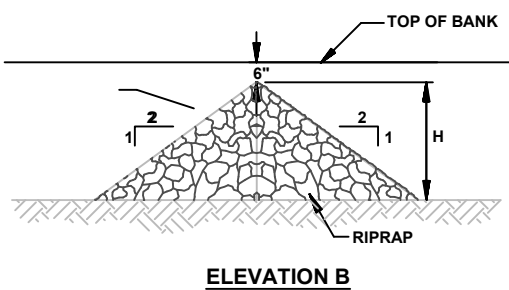
FILTER SOCK



CHECK DAMS IN SWALE

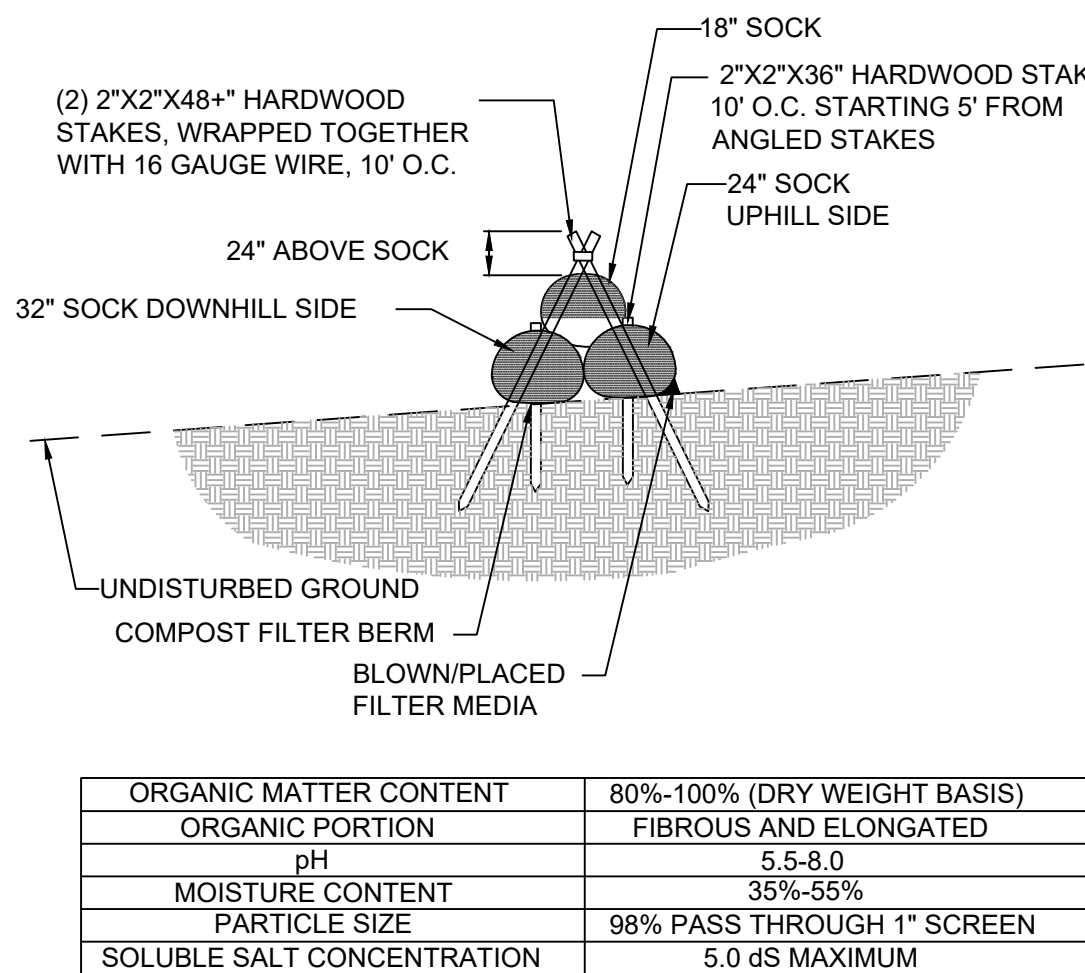
2-10 ACRES OF DRAINAGE AREA

(DOWNSTREAM VIEW)



ROCK CHECK DAM

N.T.S.



ORGANIC MATTER CONTENT	80%-100% (DRY WEIGHT BASIS)
ORGANIC PORTION	FIBROUS AND ELONGATED
pH	5.5-8.0
MOISTURE CONTENT	35%-55%
PARTICLE SIZE	98% PASS THROUGH 1" SCREEN
SOLUBLE SALT CONCENTRATION	5.0 dS MAXIMUM

MATERIAL NOTES:

1. COMPOST USED FOR FILTER SOCKS SHALL BE WEED, PATHOGEN AND INSECT FREE AND FREE OF ANY REFUSE, CONTAMINANTS OR OTHER MATERIALS TOXIC TO PLANT GROWTH. THEY SHALL BE DERIVED FROM A WELL-DECOMPOSED SOURCE OF ORGANIC MATTER AND CONSIST OF PARTICLES RANGING FROM 3/8" TO 2".

2. FILTER SOCKS SHALL BE 3 OR 5 MIL CONTINUOUS, TABULAR, HDPE 3/8" KNITTED MESH NETTING MATERIAL FILLED WITH COMPOST PASSING THE ABOVE SPECIFICATIONS FOR COMPOST PRODUCTS.

INSTALLATION NOTES:

3. FILTER SOCKS WILL BE PLACED ON A LEVEL LINE ACROSS SLOPES.

4. FILTER SOCKS INTENDED TO BE PERMANENT SHALL BE SEED AT TIME OF INSTALLATION.

5. FILTER SOCKS ARE NOT TO BE USED IN CONCENTRATED FLOW SITUATIONS OR IN RUNOFF CHANNELS.

MAINTENANCE NOTES:

6. ROUTINELY INSPECT FILTER SOCKS AFTER SIGNIFICANT RAIN , KEEPING THEM FUNCTIONAL AT ALL TIMES.

7. REMOVE SEDIMENTS COLLECTED AT THE BASE OF THE FILTER SOCKS WHEN SEDIMENT REACHES 1/3 OF THE EXPOSED HEIGHT OF THE PRACTICE.

8. WHERE THE FILTER SOCK DETERIORATES OR FAILS, IT WILL BE REPAIRED OR REPLACED WITH A MORE EFFECTIVE ALTERNATIVE.

9. FILTER SOCKS WILL BE DISPERSED ON SITE WHEN NO LONGER REQUIRED IN SUCH A WAY AS TO FACILITATE AND NOT OBSTRUCT SEEDING.

STACKING NOTES:

10. TRIPLE STACK FILTER SOCK SHALL NOT EXCEED THREE SOCKS IN HEIGHT AND SHALL BE STAKED IN PYRAMIDAL FORM AS SHOWN ABOVE. MINIMUM TRAP HEIGHT IS 24" DIAMETER SOCK.

11. PHOTODEGRADABLE AND BIODEGRADABLE SOCKS SHALL NOT BE USED FOR MORE THAN 1 YEAR.

TRIPLE STACK FILTER SOCK

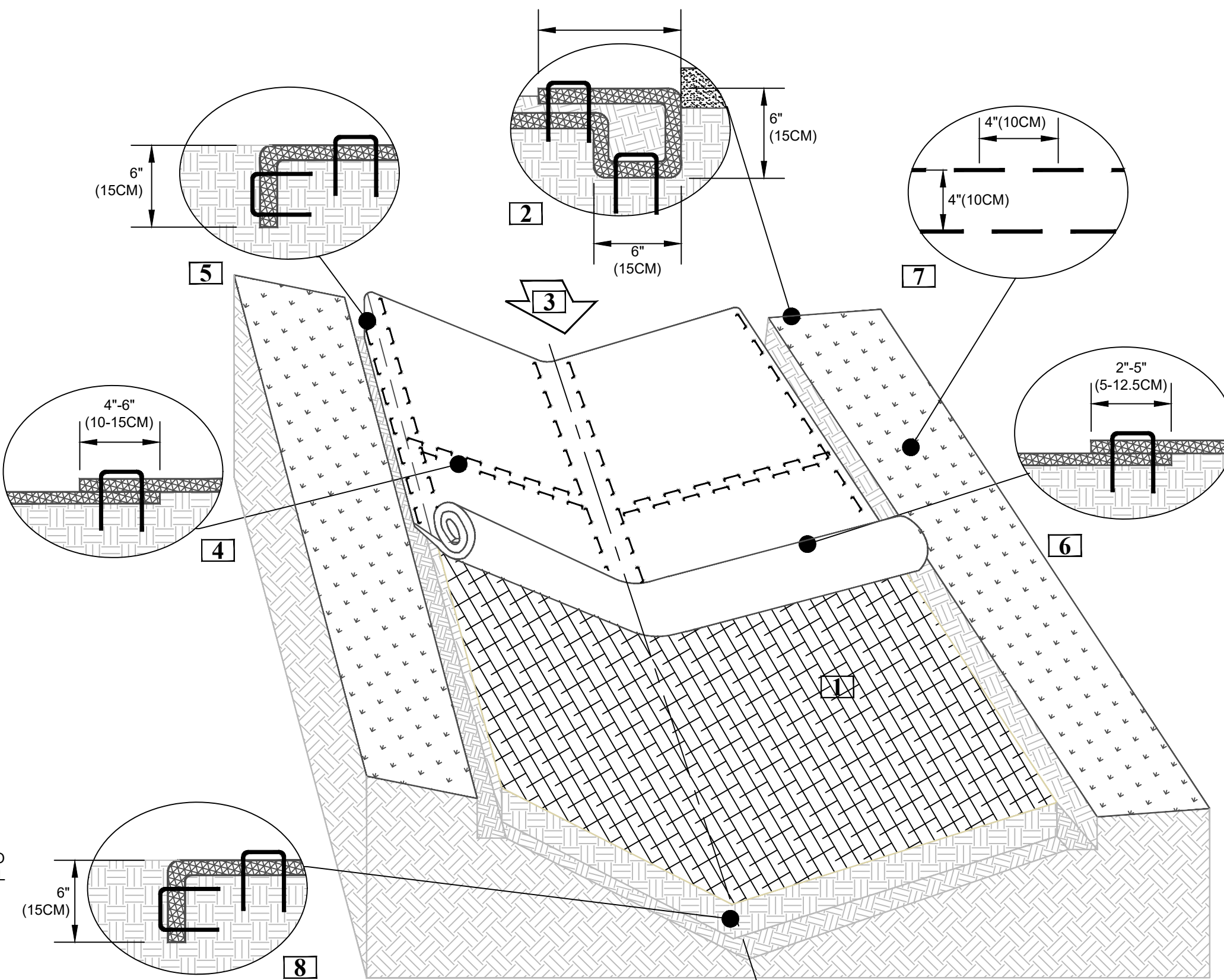
N.T.S.

1. PREPARE SOIL BEFORE INSTALLING ROLLED EROSION CONTROL PRODUCTS (RECPS), INCLUDING ANY NECESSARY APPLICATION OF LIME, FERTILIZER, AND SEED.
2. BEGIN AT THE TOP OF THE CHANNEL BY ANCHORING THE RECPS IN A 6"(15CM) DEEP X 6"(15CM) WIDE TRENCH WITH APPROXIMATELY 12"(30CM) OF RECPS EXTENDED BEYOND THE UP-SLOPE PORTION OF THE TRENCH. USE SHOREMAX MAT AT THE CHANNEL/CULVERT OUTLET AS SUPPLEMENTAL SCOUR PROTECTION AS NEEDED. ANCHOR THE RECPS WITH A ROW OF STAPLES/STAKES APPROXIMATELY 12"(30CM) APART IN THE BOTTOM OF THE TRENCH. BACKFILL AND COMPACT THE TRENCH AFTER STAPLING. APPLY SEED TO THE COMPACTED SOIL AND FOLD THE REMAINING 12"(30CM) PORTION OF RECPS BACK OVER THE SEED AND COMPACTED SOIL. SECURE RECPS OVER COMPACTED SOIL WITH A ROW OF STAPLES/STAKES SPACED APPROXIMATELY 12" APART ACROSS THE WIDTH OF THE RECPS.
3. ROLL CENTER RECPS IN DIRECTION OF WATER FLOW IN BOTTOM OF CHANNEL. RECPS WILL UNROLL WITH APPROPRIATE SIDE AGAINST THE SOIL SURFACE. ALL RECPS MUST BE SECURELY FASTENED TO SOIL SURFACE BY PLACING STAPLES/STAKES IN APPROPRIATE LOCATIONS AS SHOWN IN THE STAPLE PATTERN GUIDE.
4. PLACE CONSECUTIVE RECPS END-OVER-END (SHINGLE STYLE) WITH A 4"-6" OVERLAP. USE A DOUBLE ROW OF STAPLES STAGGERED 4" APART AND 4" ON CENTER TO SECURE RECPS.
5. FULL LENGTH EDGE OF RECPS AT TOP OF SIDE SLOPES MUST BE ANCHORED WITH A ROW OF STAPLES/STAKES APPROXIMATELY 12"(30CM) APART IN A 6"(15CM) DEEP X 6"(15CM) WIDE TRENCH. BACKFILL AND COMPACT THE TRENCH AFTER STAPLING.
6. ADJACENT RECPS MUST BE OVERLAPPED APPROXIMATELY 2'-5" (5-12.5CM) (DEPENDING ON RECPS TYPE) AND STAPLED.
7. IN HIGH FLOW CHANNEL APPLICATIONS A STAPLE CHECK SLOT IS RECOMMENDED AT 30 TO 40 FOOT (9 -12M) INTERVALS. USE A DOUBLE ROW OF STAPLES STAGGERED 4"(10CM) APART AND 4"(10CM) ON CENTER OVER ENTIRE WIDTH OF THE CHANNEL.
8. THE TERMINAL END OF THE RECPS MUST BE ANCHORED WITH A ROW OF STAPLES/STAKES APPROXIMATELY 12" (30CM) APART IN A 6"(15CM) DEEP X 6"(15CM) WIDE TRENCH. BACKFILL AND COMPACT THE TRENCH AFTER STAPLING.

NOTES:

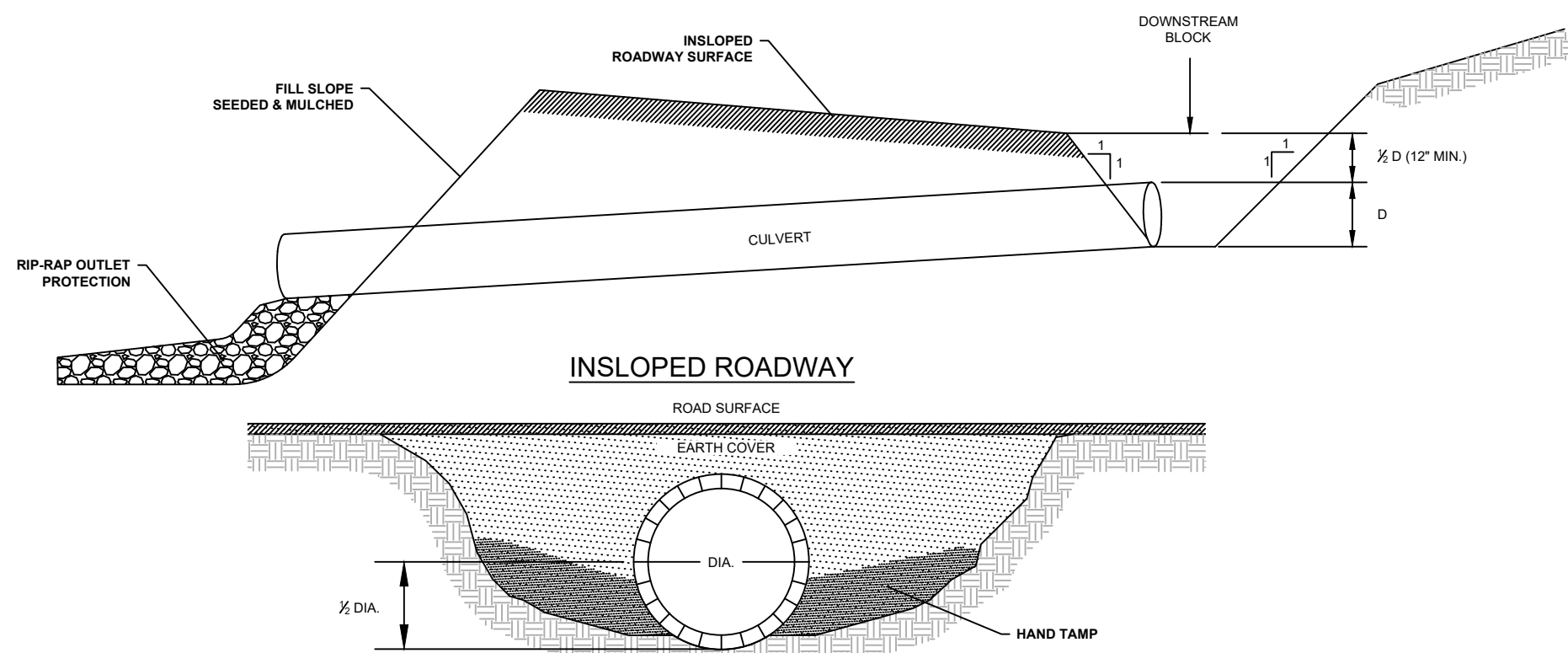
1. HORIZONTAL STAPLE SPACING SHOULD BE ALTERED IF NECESSARY TO ALLOW STAPLES TO SECURE THE CRITICAL POINTS ALONG THE CHANNEL SURFACE

2. IN LOOSE SOIL CONDITIONS, THE USE OF STAPLE OR STAKE LENGTHS GREATER THAN 6" (15CM) MAY BE NECESSARY TO PROPERLY SECURE THE RECPS.



VEGETATIVE SWALE

N.T.S.



TYPICAL CROSS DRAIN CULVERT

CUT AND FILL SLOPES SHALL BE STABILIZED IMMEDIATELY UPON COMPLETION OF ROADWAY GRADING. THESE AREAS SHALL BE BLANKETED WHEREVER THEY ARE LOCATED WITHIN 50 FEET OF A SURFACE WATER OR WITHIN 100 FEET OF A SURFACE WATER WHERE A SUITABLE VEGETATIVE FILTER STRIP DOES NOT EXIST. A DURABLE TOP DRESSING SHALL BE PROVIDED FOR SOILS HAVING A LOW STRENGTH.

ROADSIDE DITCHES SHALL BE PROVIDED WITH ADEQUATE PROTECTIVE LINING. ADEQUATELY SIZED CULVERTS OF OTHER SUITABLE CROSS DRAINS SHALL BE PROVIDED AT ALL SEEP, SPRINGS, AND DRAINAGEWAYS. DITCH RELIEF CULVERTS SHALL BE PROVIDED AT THE INTERVALS INDICATED ON THE TABLES BELOW. ROADWAY SHALL BE INSPECTED WEEKLY AND AFTER EACH RUNOFF EVENT. DAMAGED ROADWAYS, DITCHES, OR CROSS DRAINS SHALL BE REPAIRED IMMEDIATELY.

NOTES:

1. ALL CULVERTS MUST BE INSTALLED WITH NO LESS THAN 12" COVER TO THE ROAD SURFACE.
2. ALL ACCESS ROADS WILL BE LEFT IN PLACE FOR THE ACCESS TO THE FINAL POST CONSTRUCTION PADS, THEREFORE ALL ACCESS ROADS WILL BE PERMANENT.
3. ALL PERMANENT CULVERTS WILL BE NO LESS THAN 12" IN DIAMETER.
4. SEE SHEET 6 NOTE 4 FOR CULVERT TYPE AND BEDDING AND BACKFILL REQUIREMENTS.

CROSS DRAIN CULVERT



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NO. DATE

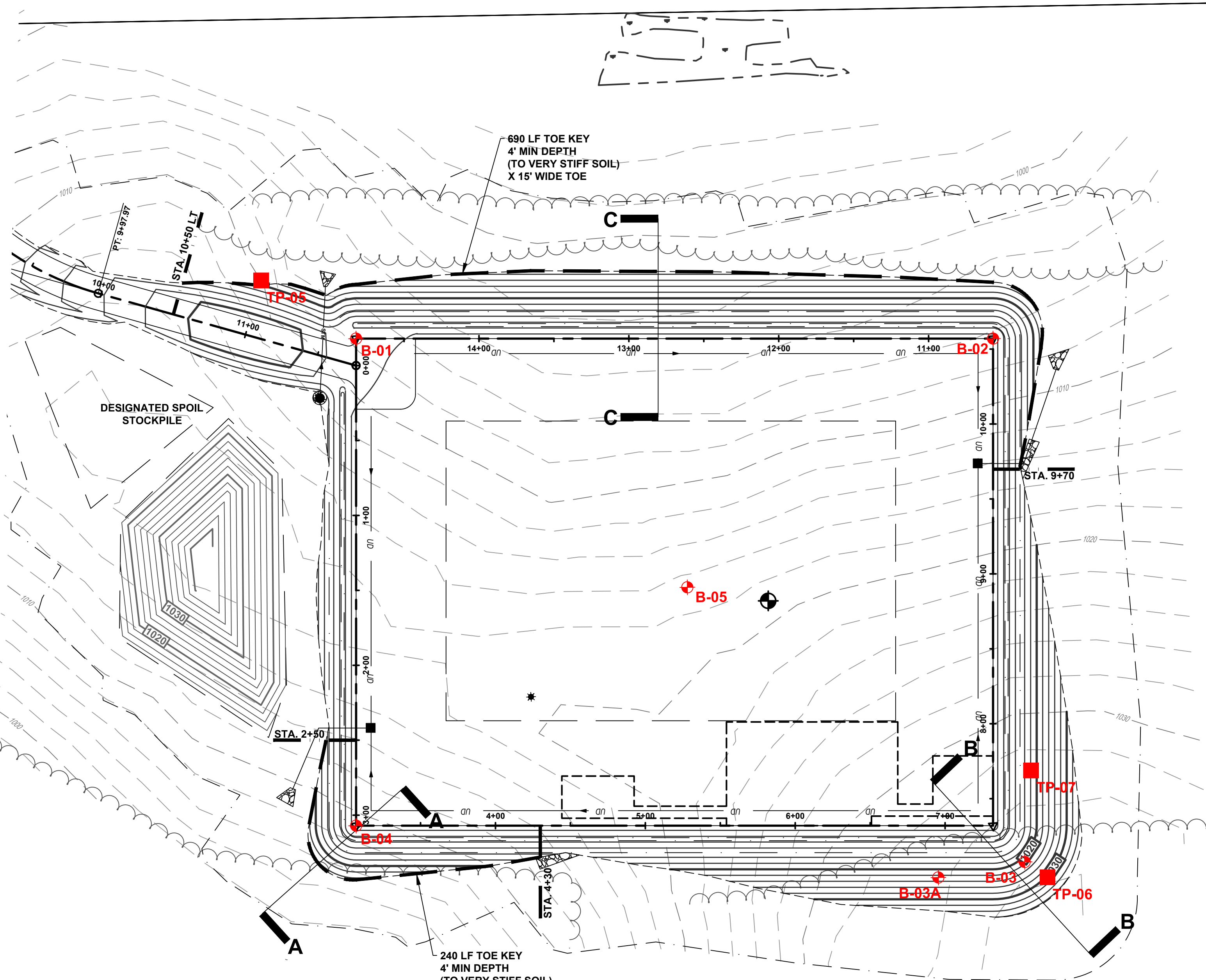


WALDO MNR GR WELL PAD

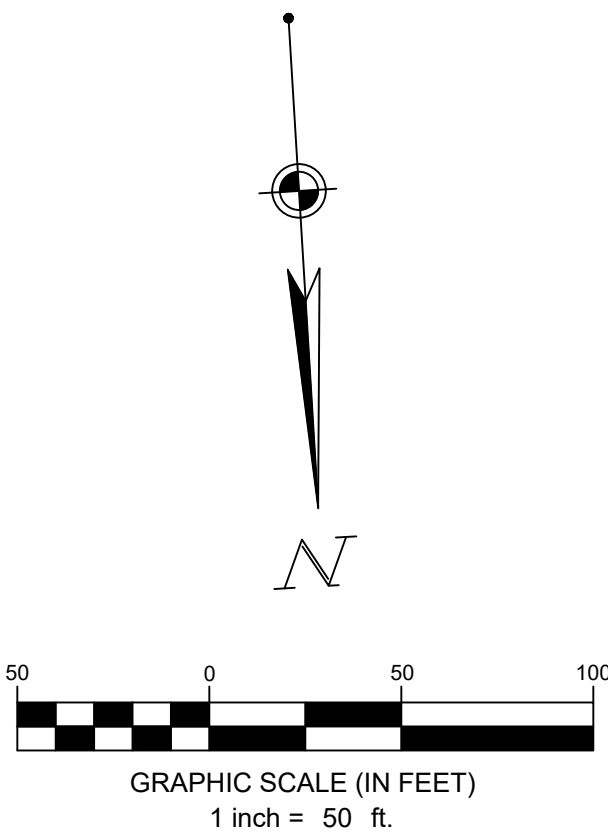
MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

EROSION & SEDIMENT CONTROL DETAILS

ISSUE:	CONST. DOCUMENTS
DATE:	08.14.2018
JOB NO.:	755215
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CHECKED:	JRM
SHEET NO.	15



MILLER WALDO C II & CARLA J
PID: 260000124000



LEGEND

EXISTING		PROPOSED	
	PROPERTY LINE		MAJOR CONTOUR
	MAJOR CONTOUR		MINOR CONTOUR
	MINOR CONTOUR		PAVEMENT
	TREE LINE		CENTER LINE
	WETLAND		UNDERDRAIN
	SOIL BORING LOCATION		STORM SEWER
	TEST PIT LOCATION		GRADE BREAK
			LIMITS OF CONSTRUCTION
			CLEARING LIMITS
			TOE KEY SECTION LIMITS
			TOE KEY, PER SHEET 17

GENERAL NOTES:

- WHEN DIRECTED BY THE ENGINEER, PERFORM TEST PITS AND PROOF-ROLLING TO EVALUATE SUBSURFACE CONDITIONS DURING CONSTRUCTION.
- CLEARING, GRUBBING, THE STRIPPING OF ORGANIC SURFACE SOILS, AND THE REMOVAL OF UNSUITABLE/UNSTABLE SOILS AND EXISTING UNCONTROLLED FILL SHALL BE PERFORMED IN ALL STRUCTURAL AREAS.
- AFTER STRIPPING HAS BEEN COMPLETED, THE RESULTING SUBGRADE SHALL BE PROOF-ROLLED WITH A FULLY LOADED TANDEM AXLE DUMP TRUCK PRIOR TO STRUCTURAL FILL PLACEMENT. SUBGRADE SOILS IDENTIFIED AS BEING UNSUITABLE OR UNSTABLE, SHALL BE UNDERCUT TO A STABLE SOIL STRATUM AND BACKFILLED WITH CONTROLLED, COMPACTED SUITABLE FILL. ALL UNSUITABLE SOIL CONTAINING EXCESSIVE ORGANICS, DEBRIS OR FROZEN MATERIAL SHALL BE WASTED AT A LOCATION APPROVED BY THE ENGINEER.
- PROVIDE DRAINAGE MEASURES FOR SPRINGS OR PERCHED WATER THAT ARE ENCOUNTERED DURING CONSTRUCTION. THE TREATMENT, WHICH WILL BE DETERMINED IN THE FIELD DEPENDING ON THE CONDITIONS, MAY INCLUDE PLACEMENT OF PERFORATED DRAIN, MATTRESS DRAINS, INTERCEPTOR TRENCHES, AND/OR SUMPS TO COLLECT WATER. ALL WATER SHALL BE OUTLETTED BY GRAVITY OUTSIDE THE LIMITS OF THE PAD SLOPES.
- THE EXCAVATION CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR EVALUATING THE EFFORT REQUIRED FOR HIGHLY WEATHERED TO COMPETENT ROCK STRATA AND TO PRODUCE EXCAVATED ROCK WITH A MAXIMUM PARTICLE SIZE OF 12 INCHES AND HAVING SUFFICIENT SOIL TO FILL VOIDS BETWEEN ROCK PARTICLES. PROVIDE ON-SITE ROCK CRUSHING AND BLENDING WITH SOIL AS NECESSARY TO MEET ACCEPTABLE GRADIATIONS FOR REUSE OF EXCAVATED ROCK AS STRUCTURAL FILL. IF EXCAVATED ROCK IS INTENDED TO BE REUSED AS ROCKFILL BY THE CONTRACTOR, IT WILL BE NECESSARY FOR THE ENGINEER TO EVALUATE THE EXCAVATED ROCK TO DETERMINE IF IT IS SUSCEPTIBLE TO SLAKING AND LOSS OF ROCK INTEGRITY. DO NOT USE SLAKING ROCK AS ROCKFILL.
- ON-SITE SOILS MEETING USCS DESIGNATIONS OF CL, ML, OR MORE GRANULAR WITH MAXIMUM PARTICLE SIZE OF SIX (6) INCHES SHALL BE USED AS SOIL FILL. UNSUITABLE SOILS, CONSISTING OF CH OR MH SOIL, OR MATERIALS CONTAINING UNACCEPTABLE FRACTIONS OF ORGANICS, DEBRIS OR FROZEN MATERIAL ARE NOT PERMITTED FOR USE AS FILL. IF COAL OR CARBONACEOUS MATERIAL IS ENCOUNTERED DURING EXCAVATION, IMMEDIATELY CONTACT THE ENGINEER.
- PLACE ALL FILL ON A STABLE, NEARLY LEVEL SUBGRADE. COMPACTION EQUIPMENT SHALL CONSIST OF LARGE (20 TON MINIMUM) VIBRATORY, SELF-PROPELLED SHEEPSFOOT ROLLERS FOR COHESIVE SOILS AND SMOOTH-DRUM ROLLERS FOR GRANULAR SOILS. SMOOTH-DRUM ROLLERS ARE NOT PERMITTED FOR THE INITIAL COMPACTION OF COHESIVE SOILS.
- COMPACT STRUCTURAL FILL TO THE FOLLOWING MINIMUM SPECIFICATIONS:
 - SOIL FILL BELOW THE UPPER TWO (2) FEET OF SUBGRADE - MAXIMUM NINE (9) INCH LOOSE LIFTS COMPACTED TO 95% OF MAXIMUM DRY DENSITY AND WITHIN THREE (3) PERCENTAGE POINTS OF THE OPTIMUM MOISTURE CONTENT AS DETERMINED BY ASTM D-698
 - SOIL FILL WITHIN THE UPPER TWO (2) FEET OF SUBGRADE - MAXIMUM NINE (9) INCH LOOSE LIFTS COMPACTED TO 98% OF MAXIMUM DRY DENSITY AND WITHIN THREE (3) PERCENTAGE POINTS OF THE OPTIMUM MOISTURE CONTENT AS DETERMINED BY ASTM D-698
 - ROCK FILL BELOW THE UPPER TWO (2) FEET OF SUBGRADE - MAXIMUM TWELVE (12) INCH LOOSE LIFTS COMPACTED TO NON-MOVEMENT UNDER COMPACTION EQUIPMENT AS APPROVED BY THE ENGINEER
 - RANDOM FILL - MAXIMUM TWELVE (12) INCH LOOSE LIFTS COMPACTED TO 90% OF MAXIMUM DRY DENSITY AND WITHIN FOUR (4) PERCENTAGE POINTS OF THE OPTIMUM MOISTURE CONTENT AS DETERMINED BY ASTM D-698
 - COMPACTION OF SOIL FILL HAVING MORE THAN 30% AGGREGATE AND ROCKFILLS SHOULD BE EVALUATED BY VISUAL OBSERVATIONS TO INCLUDE LIFT THICKNESS, GRADATION, AND MOISTURE OF THE MATERIALS, COMPACTIVE EFFORT, AND NON-MOVEMENT UNDER COMPACTION EQUIPMENT AND PROOF-ROLLING.
- IF MOISTURE CONDITIONING IS REQUIRED TO ACHIEVE ADEQUATE COMPACTION, CHEMICAL DRYING WITH LIME OR PORTLAND CEMENT IS ACCEPTABLE.
- FILL SLOPES SHALL BE CONSTRUCTED IN A MANNER TO ENSURE ADEQUATE COMPACTION TO THE EDGE OF THE FILL, WHICH MAY REQUIRE CONSTRUCTING THE SLOPES WIDER THAN NECESSARY THEN CUTTING BACK TO THE REQUIRED GRADE. UPON COMPLETION OF THE FINAL GRADING, THE SLOPES SHALL BE TRACKED DOWN WITH TRACKED EQUIPMENT PRIOR TO PLACING TOPSOIL.
- INSTALL TOE AND BONDING BENCHES AS NOTED ON THIS PLAN. DIFFERENT BENCH DIMENSIONS AND CONFIGURATIONS MAY BE NECESSARY AS DICTATED BY SUBSURFACE CONDITIONS, SUCH AS WET OR UNSTABLE SOILS, HIGH PLASTICITY CLAYS, OR OTHER WEAK MATERIALS. BENCH DEPTHS MAY BE MODIFIED IN THE FIELD BY THE ENGINEER DEPENDING ON ACTUAL CONDITIONS. SHALLOWER DEPTHS MAY BE APPROVED BY THE ENGINEER IF STABLE HIGHLY WEATHERED ROCK IS ENCOUNTERED. GREATER DEPTHS MAY BE DIRECTED BY THE ENGINEER IF SUBGRADE SOILS ARE NOT SUITABLE FOR SUPPORT OF SLOPE. THESE OCCURRENCES MAY BE REVEALED DURING GRADING, OR DURING TEST PIT EXCAVATION PERFORMED DURING TOE BENCH CONSTRUCTION. DRAINAGE MEASURES SHALL BE PROVIDED IN THE TOE BENCHES AS INDICATED ON THIS PLAN. ADDITIONAL DRAINAGE MEASURES SHALL BE INSTALLED AS DIRECTED BY THE FIELD ENGINEER. BENCH DRAINS SHALL CONNECT TO DRAINAGE MAINS, WHICH ARE OUTLETTED VIA GRAVITY LATERALS WHERE INDICATED.
- WHERE SLOPES ARE PLACED OVER EXISTING SLOPES 8H:1V OR STEEPER CONSTRUCT BONDING BENCHES TO THE LIMITS INDICATED ON THE PLANS AND CROSS SECTIONS. BONDING BENCHES SHALL BE CONSTRUCTED TO HAVE A MINIMUM EMBEDMENT BELOW THE NATIVE GROUND SURFACE AS INDICATED ON THIS PLAN.
- DRAINAGE MEASURES SHALL BE PROVIDED ON THE TOE AND BONDING BENCHES AS INDICATED ON THE PLANS. LOCATIONS OF COLLECTOR/OUTLET ARE SCHEMATIC AND ALTERNATE LOCATIONS CAN BE USED AS LONG AS GRAVITY DRAINAGE IS PROVIDED. ALL DRAINS (TOE, TRENCH AND BENCH) SHALL HAVE A 4 INCH (MIN) PERFORATED PIPE AND BE BACKFILLED WITH AASHTO NO. 57 AGGREGATE WRAPPED IN A SEPARATION FILTER FABRIC OR AN APPROVED ALTERNATE DRAIN. DRAINS SHALL HAVE A MINIMUM 2% GRADE LATERALLY TO DRAIN TO COLLECTOR/OUTLET LOCATIONS. PROVIDE A 6 INCH (MIN) SOLID (I.E. NON-PERFORATED) COLLECTOR/OUTLET PIPE TO DISCHARGE THE DRAINS BEYOND THE TOE OF SLOPE. PROVIDE RIPRAP PROTECTION AT THE OUTLET LOCATIONS TO MINIMIZE SURFACE EROSION. CHANGES TO THE CONFIGURATION AND NUMBER OF DRAINS MAY BE MADE WITH THE ENGINEER'S APPROVAL BASED UPON WATER CONDITIONS ENCOUNTERED DURING BENCHING.
- THE SIDES OF TEMPORARY EXCAVATIONS SHALL BE ADEQUATELY SLOPED OR SHORED/BRACED TO PROVIDE STABLE SIDES AND SAFE WORKING CONDITIONS FOR OPERATORS, LABORERS AND ENGINEERING INSPECTORS TO PROOFROLL THE SUBGRADE AND PERFORM TEST PITS. ALL APPLICABLE LOCAL, STATE, FEDERAL, AND OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA) SAFETY STANDARDS SHALL BE FOLLOWED BY THE CONTRACTOR.
- PROOF-ROLL THE FINAL PAD/ROAD SURFACE WITH A LOADED TRIAXIAL DUMP TRUCK TO IDENTIFY ANY AREAS OF UNSTABLE SOILS. UNSTABLE SOILS SHALL BE REMOVED BY OVER-EXCAVATION AND REPLACEMENT WITH CONTROLLED COMPACTED SOIL FILL.
- THE SUBGRADE SHALL BE SEALED ON A DAILY BASIS TO MINIMIZE DISTURBANCE AND MOISTURE INTRUSION.
- THE OBSERVATION AND TESTING SHALL INCLUDE OBSERVATION OF STABILITY OF TOE AND BONDING BENCHES, INSTALLATION OF DRAINAGE MEASURES, LIFT THICKNESS OF MATERIAL, COMPACTIVE EFFORT, TYPE AND MOISTURE OF SOIL, GRADATION OF THE ROCK FILL, AND MOISTURE AND DENSITY OF COMPACTION. ALL COMPACTIVE EFFORT SHALL BE VERIFIED BY IN-PLACE DENSITY TESTING OR VISUAL OBSERVATION/PROOF-ROLLING FOR FILLS HAVING MORE THAN 30% AGGREGATE (>3/4") PARTICLE SIZES.



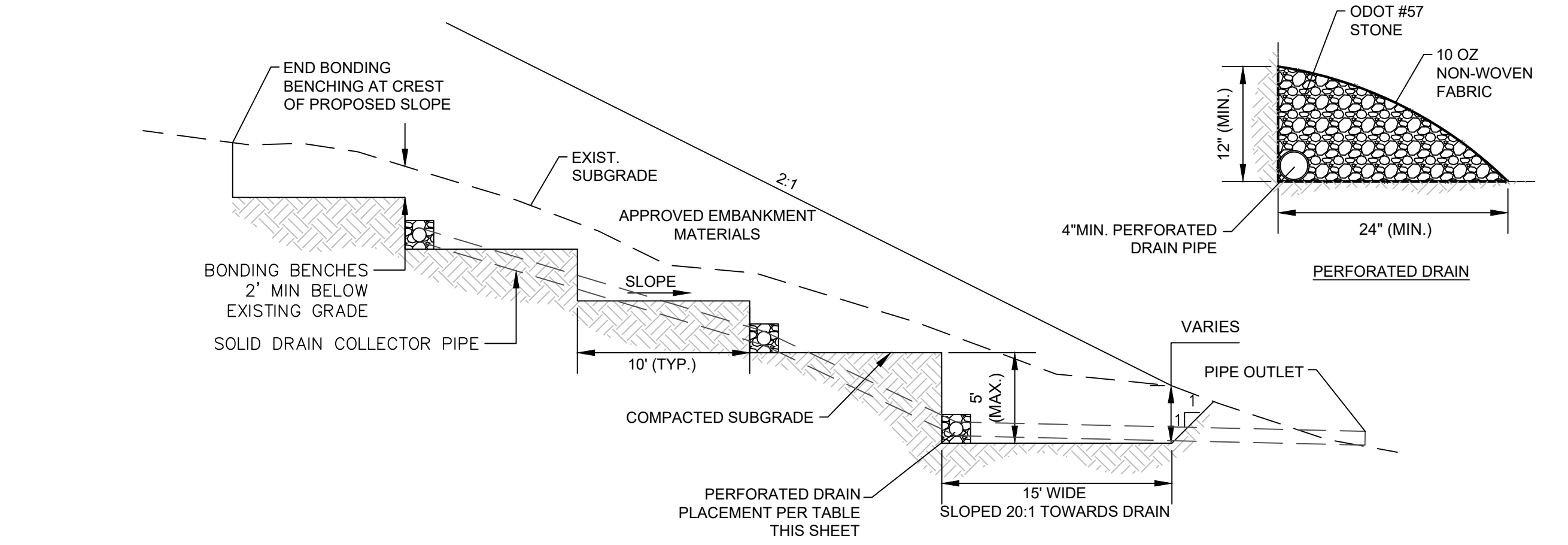
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WALDO MNR GR WELL PAD
MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

GEOTECHNICAL
ENGINEERING
PLAN

ISSUE:	CONST. DOCUMENTS
DATE:	08.14.2018
JOB NO.:	755215
DESIGN:	SLE
DRAWN:	JMR
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SHEET NO.	16



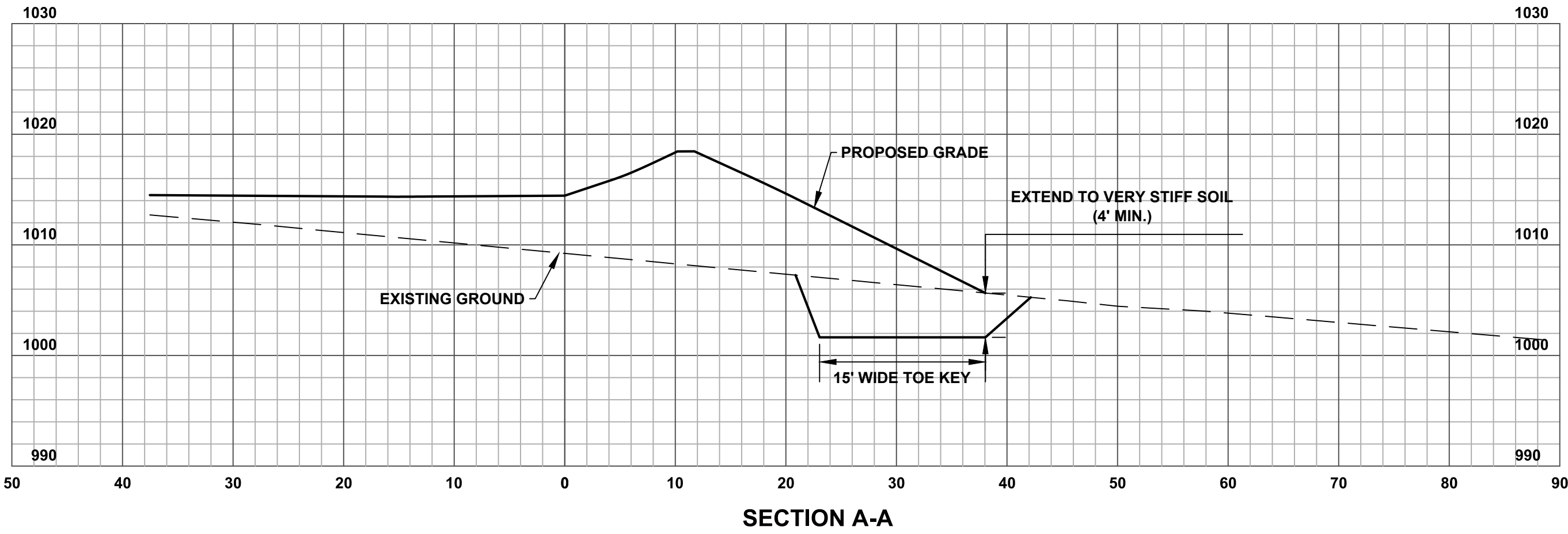
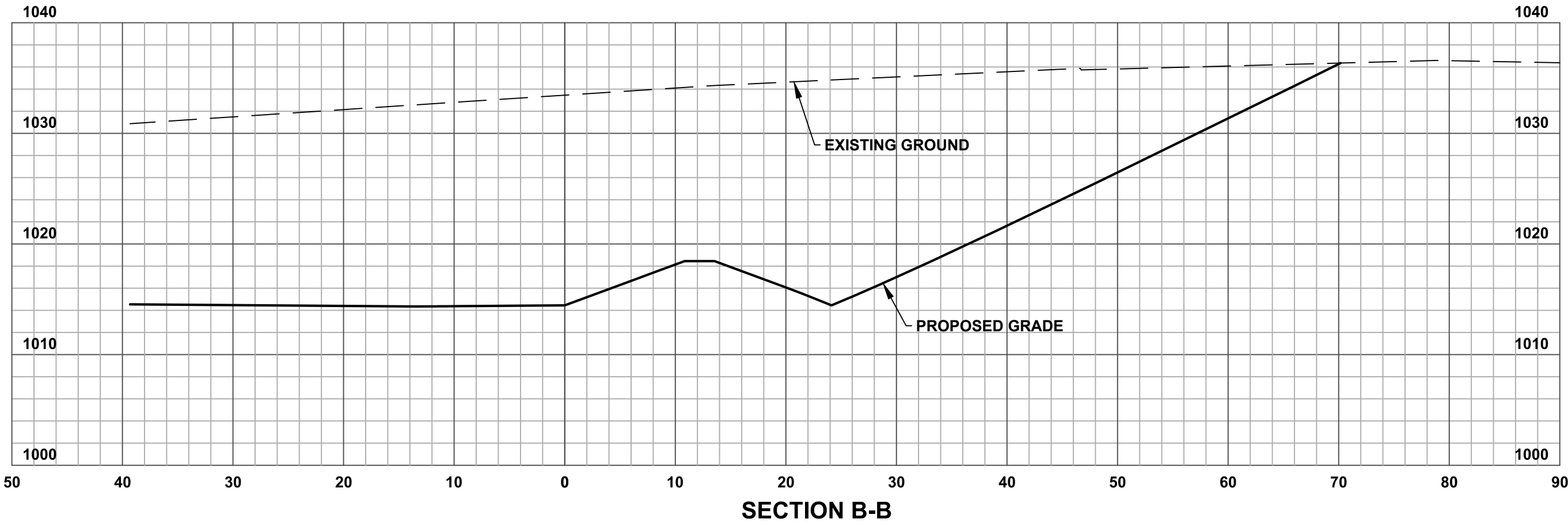
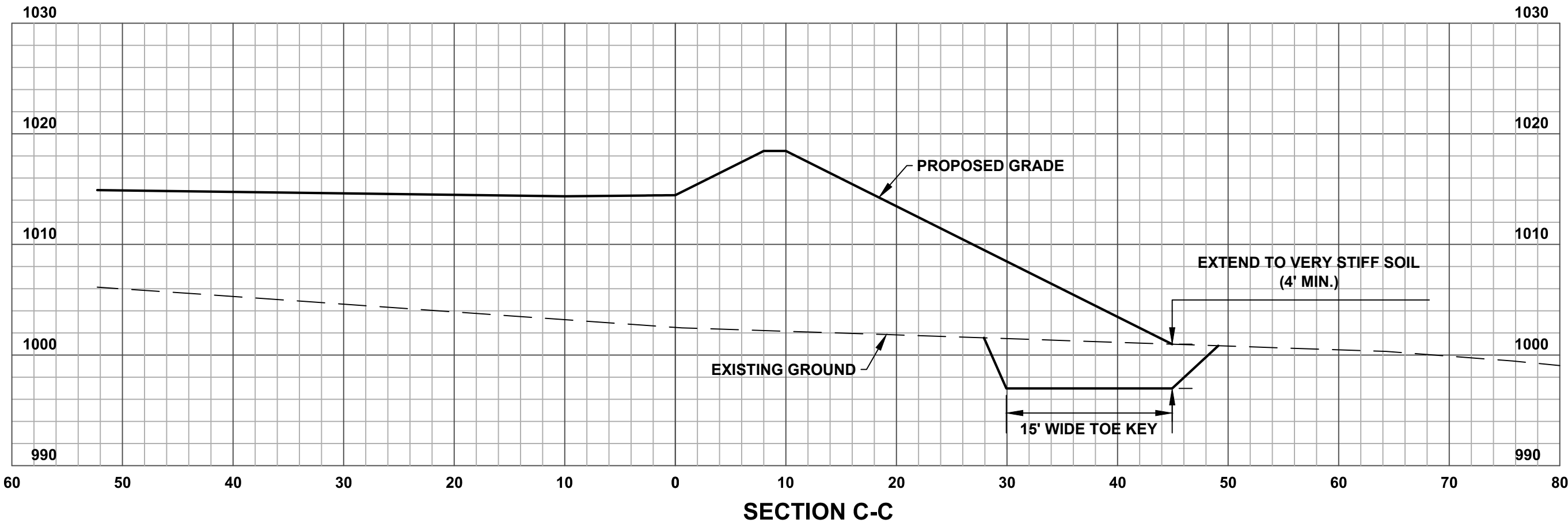
- NOTES:
- TOE BENCHES SHALL BE CONSTRUCTED IN COMPETENT MATERIAL TO THE MINIMUM DEPTH REQUIRED. ADDITIONAL DEPTHS MAY BE REQUIRED IN THE FIELD DEPENDING UPON THE STABILITY OF SUBGRADE MATERIAL. AT THE REQUEST OF THE FIELD ENGINEER, PERFORM TEST PITS AND PROOFROLLING OF THE BENCH SUBGRADE TO EVALUATE THE SUBGRADE CONDITIONS. REMOVE UNSTABLE SOILS AND EXTEND THE DEPTH OF THE TOE BENCH AS REQUESTED BY THE ENGINEER. PROVIDE A HORIZONTAL (ALONG THE LENGTH OF THE TOE BENCH) GRADE OF AT LEAST 2% TO ALLOW WATER TO DRAIN TO LOW POINTS FOR OUTLET.
 - INSTALL PERFORATED BENCH DRAINS IN ACCORDANCE WITH THIS DETAIL. PROVIDE HORIZONTAL GRADE OF 2% TO ALLOW WATER TO DRAIN TO OUTLET.
 - BONDING BENCHES SHALL BE EMBEDDED A MINIMUM OF TWO (2) FEET BELOW THE NATIVE/ORIGINAL GROUND SURFACE EXCEPT WHERE NOTED. ADJUST THE HEIGHT OF THE BONDING BENCH TO ACCOMMODATE THE TWO (2) FOOT MINIMUM EMBEDMENT DEPTH.
 - PROVIDE A 20H:1V SLOPE ON ALL BENCH BOTTOMS IN THE DIRECTION SHOWN TO DIRECT SEEPAGE INTO PERFORATED DRAINS PLACED AT THE REAR OF THE BENCH.
 - TERMINATE BONDING BENCHES WHEN THE NATIVE SLOPES ARE FLATTER THAN 8H:1V OR A MINIMUM OF TEN (10) FEET BEHIND THE CREST OF SLOPE.
 - THE CONTRACTOR IS RESPONSIBLE FOR SAFETY OF EXCAVATIONS, INCLUDING SEQUENCING OF BENCHING OPERATIONS, AND SLOPES PER OSHA 29CFR 1926.650 REQUIREMENTS, TO ALLOW THE APPROPRIATE ENGINEERING EVALUATIONS. THIS MAY REQUIRE STARTING EXCAVATION AT THE MID-HEIGHT OF THE BONDING BENCHES TO REMOVE THE SLOPE OVERBURDEN THAT COULD SLIDE INTO THE SUBSEQUENT TOE BENCH EXCAVATION.
 - BACKFILL THE BOTTOM TWO BENCHES WITH ON-SITE ROCK FILL, IF MATERIAL IS AVAILABLE

TYPICAL SECTION - EMBANKMENT BENCHING & TOE KEY
N.T.S.

SUMMARY OF SLOPE STABILIZATION MEASURES

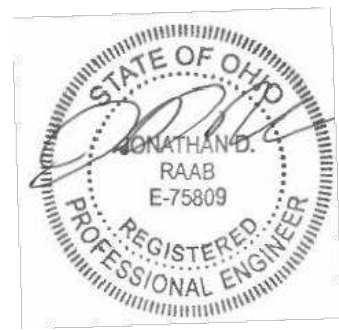
SLOPE STATIONING	PROPOSED SLOPE (H:V)	*MAX SLOPE HEIGHT (FT.)	**MIN TOE BENCH DIMENSIONS (FT.)	***DRAINAGE MEASURES REQUIRED	****DRAINAGE OUTLET LOCATIONS	ADDITIONAL TOE BENCH LEVELS	##BONDING BENCHES REQUIRED
WELL PAD							
2+50 TO 4+30	2H:1V	12	EXTEND 4 FT MIN (TO VERY STIFF SOIL)	NONE	NONE	NONE	NONE
9+70 TO 10+50 LT	2H:1V	17	EXTEND 4 FT MIN (TO VERY STIFF SOIL)	NONE	NONE	NONE	NONE

* MEASURED FROM TOE TO CREST
** BENCH DEPTHS MAY BE MODIFIED IN THE FIELD BY THE ENGINEER DEPENDING ON THE ACTUAL CONDITIONS. DESCRIPTIONS OF REQUIRED SOIL CONSISTENCY (I.E. STIFF, VERY STIFF, HARD) OR DENSITY (I.E. MEDIUM DENSE, DENSE, VERY DENSE) ARE BASED ODOT'S SPECIFICATIONS FOR GEOTECHNICAL EXPLORATIONS TABLES 600-1 AND 600-2, AND ARE ALSO SHOWN IN APPENDIX B "NOTES FOR EXPLORATION LOGS". SHALLOWER DEPTHS CAN BE USED IF STABLE HIGHLY WEATHERED ROCK WITH A CONTINUOUS DEPTH IS ENCOUNTERED. GREATER DEPTHS MAY BE NECESSARY IF SUBGRADE SOILS ARE NOT STABLE FOR SUPPORT OF SLOPE. IN AREAS WHERE BENCHING IS NOT SHOWN, INSTALL BENCHING WITH MINIMUM DEPTHS IN ACCORDANCE WITH CURRENT ODOT BENCHING STANDARDS.
*** PROVIDE A 4-INCH DIAMETER (MIN) PERFORATED SCHEDULE 40 DRAIN PIPE AND BACKFILL WITH AASHTO NO. 57 AGGREGATE WRAPPED IN FILTER FABRIC OR APPROVED ALTERNATE DRAIN. DRAIN SHALL HAVE A MINIMUM 2% GRADE LATERALLY TO DRAIN TO OUTLET LOCATION.
****LOCATIONS OF COLLECTOR/OUTLET DRAINS ARE SCHEMATIC AND ALTERNATE LOCATIONS CAN BE USED AS LONG AS GRAVITY DRAINAGE IS PROVIDED. DRAINS SHALL HAVE A MINIMUM 2% GRADE LATERALLY TO DRAIN TO OUTLET LOCATION. PROVIDE 6-INCH DIAMETER (MIN) SOLID (I.E. NON-PERFORATED) SCHEDULE 40 COLLECTOR/OUTLET PIPE TO DISCHARGE THE DRAINS BEYOND THE TOE OF SLOPE. PROVIDE RIPRAP PROTECTION AT THE OUTLET LOCATIONS TO MINIMIZE SURFACE EROSION.
ALL SLOPES, INCLUDING SLOPES NOT LISTED IN THIS TABLE, THAT ARE CONSTRUCTED OVER EXISTING SLOPES STEEPER THAN 8H:1V SHALL HAVE A BONDING BENCHING IN ACCORDANCE WITH PROJECT DETAILS. BONDING BENCH DIMENSIONS SHOWN ARE APPROXIMATE. DIFFERENT BENCH DIMENSIONS ARE PERMITTED AS LONG AS THE ACTUAL BENCHING FOLLOWS THE "AVERAGE" BENCH EXCAVATION CONFIGURATION DEPICTED.



REVISION DESCRIPTION

NO. DATE



WALDO MNR GR WELL PAD

MONROE TOWNSHIP, GUERNSEY COUNTY, OHIO

GEOTECHNICAL
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SECTIONS

ISSUE:
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17